Search Frictions and Welfare in a Market for Prominence: Theory and Experiment*

Zitian Wang†

November 7, 2026

Click here for the latest draft

Click here for the latest slides

Abstract

In online marketplaces, sellers pay platforms to make their products more visible and easier to inspect, creating a market for search prominence. I study how consumer search friction affects the level and distribution of surplus in this market. In the model, two sellers bid for a prominent position in a second-price auction and then post prices. The consumer observes both prices and learns her match value for the prominent seller's product for free before deciding whether to incur a search cost to inspect the non-prominent seller's product. I test the model's predictions in a laboratory experiment that exogenously varies search cost between low and high levels. I find that platform revenue is insensitive to a doubling of search cost, despite the predicted doubling of revenue. Consumer surplus exceeds predicted levels and accounts for a larger share of total surplus than predicted in both treatments. Sellers jointly retain 20–22 percent of total surplus after paying for prominence. The estimated agent quantal response equilibrium captures the observed insensitivity of bids and compressed price gap that explain these welfare findings.

JEL Classification Number: C72, C91, D83

Keywords: consumer search, paid prominence, price competition, welfare, experiment

*Financial support from the Director of Graduate Studies in Economics and the Graduate School of Arts and Sciences Council at the University of Virginia is gratefully acknowledged. I am especially grateful to my advisors Charles Holt, Po-Hsuan Lin, and Peter Troyan for their guidance. I am also grateful to Simon Anderson for his valuable advice on this project. I thank Nageeb Ali, Felipe Araujo, Ala Avoyan, Tony Fan, Meng-Jhang Fong, Menglong Guan, Paul Healy, Shengwu Li, Yuval Lidany, Suraj Malladi, Alessandro Pavan, Jiarui (Jerry) Qian, Zhongheng Qiao, Emanuel Vespa, and Joseph Tao-yi Wang for their insightful comments. I also thank my colleagues in the UVA Theory/Experimental Economics Workshop, including Istiak Ahmed, Selena Cheung, William Donnelly, Tanvi Goyal, Yijun Shen, Yuchen Song, Patarasate Unjitwattana, Pin-Chao Wang, and Emily Yunxi Xie, for helpful discussions. I am grateful to Daniel Kwiatkowski and Yen Ling Tan for their advice during my graduate studies. I thank participants at BEEMA10, the Northwestern-Kellogg Summer School in Economic Theory, the 2026 ESA World and North American Meetings, the 2026 C-BID Job Market Workshop at NYU Abu Dhabi, and the UVA Computer Science Theory Seminar for their feedback. The experiment is approved by the University of Virginia IRB #8298. All errors are mine.

†University of Virginia, Department of Economics. Email: zw3qt@virginia.edu.

1 Introduction

Online marketplaces offer a wide range of products, but these products are not displayed equally prominently. Some appear at the top of search results or receive more visible display space, while others require consumers to scroll through several pages of results. The FTC's complaint against Amazon states that "70% of Amazon shoppers do not click past the first Search Results Page."[1] These differences in visibility can affect which products consumers consider and ultimately buy. Sellers therefore pay platforms to make their products easier to discover and inspect, creating a market for search prominence. Examples include Google's search ads, Amazon's Sponsored Products, and paid hotel listings on Expedia. These forms of advertising have reshaped the landscape of competition in online markets and generated substantial platform revenue. In 2025, Alphabet reported $224.5 billion in "Google Search & other" advertising revenue.[2]

Costly consumer search gives sellers a reason to compete for prominence. In physical markets, search can involve traveling between stores to obtain product information. Online shopping reduces these physical costs, but consumers still spend time and effort acquiring detailed information on product characteristics and assessing whether a product suits their needs and tastes. A more prominent product can therefore attract purchases from consumers who stop before considering other products. Consumers' search and purchase decisions shape sellers' pricing incentives, while the prices consumers encounter affect whether they continue searching. These strategic interactions determine the additional revenue associated with a more prominent position and hence sellers' willingness to pay for prominence.

This paper studies how search cost affects the level and distribution of surplus in a market for prominence. I develop a model in which sellers compete for prominence through an auction and then compete through pricing in the product market. In a financially incentivized laboratory experiment, I vary search cost exogenously between low and high levels to test predictions for the division of surplus among consumers, sellers, and the platform. In both treatments, I find that consumers receive more surplus than equilibrium predicts, while sellers jointly retain only about 20 percent of total surplus after paying for prominence. I use observed bidding, pricing, and search behavior to explain these and other findings.

The expansion of online commerce has made price information more accessible. For example, prices appear in product listings and on price comparison websites, allowing consumers to compare them without detailed product inspection. An important implication of visible prices is that sellers can influence consumers' search decisions through their pricing strategies. A low price for a product already examined can encourage an immediate purchase without further search. A low price for a product not yet examined, however, can encourage

[1] See the Federal Trade Commission's Second Amended Complaint against Amazon, paragraph 97.
[2] Alphabet Inc., 2025 Annual Report, "Google Search & other" advertising revenue.

further search by increasing the expected gain from doing so. Prices thus affect not only how the surplus from a transaction is divided between consumers and sellers, but also which products consumers examine.

In the model, two *ex ante* identical sellers compete for a prominent position in a second-price auction and then simultaneously post prices to compete for sales in the product market. Each seller offers one horizontally differentiated product to a buyer with unit demand. Each product can be a good or bad match for the buyer, and match values are independent across products. The buyer observes both prices and learns her match value for the prominent seller's product for free. She then decides whether to pay a search cost to learn her initially hidden match value for the non-prominent seller's product. At any point, she may buy a product she has already inspected or exit without purchasing. When bidding for the prominent position, each seller anticipates how winning or losing the auction would affect both sellers' prices and the buyer's search and purchase decisions. A seller's willingness to pay for the prominent position equals the seller's expected revenue from the product market in the prominent position minus the expected revenue in the non-prominent position.

Identifying the welfare effects of search cost in field data is challenging for two reasons. The first is finding exogenous variation in search cost. Platforms may adjust search costs in response to market conditions that researchers do not observe and that also affect prices, purchases, and welfare. This makes it difficult to separate the effect of search cost from the effects of the conditions that motivated the platform's decision to change search cost. The second is that prominence in the field can combine a search-cost advantage with signals of quality or relevance. Consumers may interpret a high ranking as a personalized recommendation of a product that suits their tastes. They may also regard a seller's willingness to pay for prominence as a signal of higher product quality. These interpretations make it difficult to isolate the effect of the search-cost advantage itself.

I address these challenges by implementing the model in a between-subjects laboratory experiment that exogenously assigns low or high search cost, while holding other features of the market fixed. Because match values and search costs are assigned in the experiment, I can calculate each party's surplus from participants' search decisions and transactions. I choose the experimental parameters to generate distinct predictions across the two treatments. Under these parameter values, the model predicts that 1) sellers' bids and platform auction revenue rise one for one with search cost, 2) higher search cost reduces total surplus, consumer surplus, and combined seller profits, 3) both sellers use mixed pricing strategies, with the prominent seller's price distribution first-order stochastically dominating the non-prominent seller's, 4) search follows a cutoff determined by both prices and the buyer's match with the prominent product, with further search becoming less frequent at the higher search cost, and 5) sellers jointly receive the largest share of total surplus at the low search cost and

the platform receives the largest share at the high search cost, while consumers receive the smallest share in both treatments.

In the experiment, I find that consumer surplus exceeds equilibrium predictions in both treatments and falls proportionally much less as search cost increases. These findings are accompanied by substantial departures from equilibrium pricing. Prominent sellers charge higher prices on average than non-prominent sellers, as predicted, but prominent sellers' average prices remain substantially below their equilibrium benchmarks. Non-prominent sellers charge more than predicted at the high search cost. The average price gap between positions is compressed relative to the equilibrium prediction in both treatments. Prices at both positions are also more dispersed than predicted and extend beyond the predicted price ranges. When the prominent product is a good match, lower prominent prices increase the surplus from buying it and can make further search unnecessary, saving the buyer the search cost. When the prominent product is a bad match, however, high non-prominent prices can discourage further search and thus reduce demand for the non-prominent seller.

I also find that sellers jointly retain only about 20 percent of total surplus as profit in both treatments. This small share has two sources. Sellers earn less revenue from the product market than predicted in both treatments, while payments for the prominent position, determined by losing bids, absorb a substantial share of that revenue. On average, bids exceed the realized revenue advantage of the prominent position in both treatments and the equilibrium bidding benchmark at the low search cost, but remain close to that benchmark at the high search cost. The platform receives the largest share in both treatments, while consumers receive more than the two sellers combined, in contrast to the equilibrium prediction.

Platform auction revenue changes little between the two treatments, despite the predicted increase. This is because bids rise with search cost by much less than predicted, and losing bids barely change. The platform's share of total surplus increases, but by much less than predicted. Lower search cost therefore increases consumer and total surplus in the experiment without substantially reducing platform auction revenue. The model's predicted conflict between auction revenue and consumer welfare turns out to be much less pronounced in the data.

To further explain the observed insensitivity of bids to search cost and the compressed price gap, I estimate the agent quantal response equilibrium (AQRE), in which choices respond to expected payoff differences less sharply than under best responses and players anticipate noise in subsequent decisions. With one common precision parameter estimated across decision stages and treatments, the fitted AQRE closely reproduces both patterns.

This paper contributes a theoretical and experimental framework for studying how a market friction imposed on consumers shapes sellers' competition for prominence and the distribution of surplus. I model an auction for prominence followed by a product market

in which consumers search to learn their product match values and sellers' prices influence their search decisions. The value of prominence therefore arises from the strategic interaction between sellers and buyers, and the auction determines how much sellers pay the platform for that search-cost advantage. Exogenous variation in search cost identifies its causal effects on welfare, and direct observation of match values and participants' decisions in the laboratory allows me to calculate precisely how much surplus each party receives. Evidence on bidding, pricing, and search behavior explains how these welfare outcomes arise.

1.1 Related literature

This paper relates to four strands of literature.

First, the paper relates to the literature on consumer search and price competition. Diamond (1971) shows that costly search can sustain monopoly pricing when products are homogeneous and prices are hidden until consumers search. Building on Wolinsky (1986), Anderson and Renault (1999) study differentiated products and show that higher search costs raise equilibrium prices when consumers search to learn both prices and product matches. However, this positive relationship between search cost and prices can reverse when prices are visible before search. Choi et al. (2018) and Obradovits and Plaickner (2023) show that lower search costs can raise equilibrium prices and seller profits while reducing consumer surplus. When search is more costly, a seller that attracts an early visit is more likely to make a sale before consumers consider its rivals. Sellers therefore compete more aggressively through their publicly observed prices to attract these visits. Whether prices are hidden or visible before search thus matters for how search cost affects competition and welfare.

Second, the paper relates to the literature on prominence and ordered search. Armstrong et al. (2009) exogenously designate a prominent firm that consumers visit first if they search, while Zhou (2011) takes the entire search order as exogenously given. Under hidden prices, these models show that firms searched earlier can charge lower prices than firms searched later. Earlier firms have a larger share of price-sensitive demand from consumers who can be induced to stop searching and buy immediately. Armstrong et al. (2009) also show that higher search costs raise prices at both prominent and non-prominent firms. Complementing this theoretical work, Ursu (2018) uses randomized hotel rankings on Expedia to show that rankings affect which hotels consumers click on, but have no detectable effect on purchases conditional on clicking.

Third, the paper contributes to the literature on markets for prominence and sponsored-search auctions. Edelman et al. (2007) and Varian (2007) analyze position auctions taking click volumes and advertisers' values per click as given, without explicitly modeling consumers' search decisions. Chen and He (2011) and Athey and Ellison (2011) incorporate consumer search, allowing rankings to convey information about advertisers' relevance and thereby affect the value of positions. Armstrong and Zhou (2011) show that price competition in the

product market also determines firms' willingness to pay for prominence. Under hidden prices, Anderson and Renault (2026) show that a position's value reflects two search externalities, the demand lost to firms searched earlier and the price competition induced by firms searched later. Xu et al. (2011) show that the seller with lower production costs does not always win prominence, because bids reflect how positions affect subsequent price competition. My model considers *ex ante* identical sellers, so the auction does not sort firms by production efficiency. Prices are visible before search, allowing the non-prominent seller to lower its price to encourage the buyer to search its product. I examine how search cost changes this pricing interaction and the expected revenue advantage that determines bids for prominence.

Lastly, the paper contributes to experimental studies of consumer search and market outcomes. Davis and Holt (1996) find that higher search costs raise transaction prices, but monopoly pricing is not generally observed. In contrast, Abrams et al. (2000) find no statistically significant price response to higher search cost, consistent with their model's prediction that equilibrium prices remain unchanged across the two positive search costs. The division of surplus can nevertheless depart substantially from equilibrium predictions. In treatments with human buyers where equilibrium assigns buyers little or no surplus, Abrams et al. (2000) and Cason and Friedman (2003) observe prices below the predicted levels, leaving buyers a larger share of transaction surplus. Vragov et al. (2019) study a sponsored-search market in which sellers bid for search positions and set product prices, while buyers differ in search cost. They report more search and more frequent efficient allocations among buyers with lower search costs. My experiment instead varies search cost across otherwise identical markets, to examine how sellers adjust their bids and prices. I use these responses, together with buyers' search decisions, to explain the level and distribution of surplus and their departures from equilibrium predictions.

Section 2 presents the theoretical framework. Section 3 describes the experimental design and hypotheses. Section 4 reports the welfare and behavioral results. Section 5 examines how consumer search responds to prices and the match with the prominent product. Section 6 evaluates whether the AQRE can account for bids and prices responding less to search cost than the standard equilibrium predicts. Section 7 concludes. Proofs, welfare calculations, experimental instructions, and the experimental interface are relegated to the appendices.

2 Theoretical Framework

2.1 Model

There are two ex-ante identical sellers and one buyer with unit demand. Each seller offers one product at zero marginal cost. The buyer's match values are independent across sellers, with

$$v_k = \begin{cases} V & \text{with probability } q \\ 0 & \text{with probability } 1-q \end{cases} \qquad k = 1, 2 \tag{1}$$

where $V > 0$ and $0 < q < 1$. The cost of searching the second seller's product satisfies $0 < s < qV$. The timing, information structure, payoff functions, match distribution, and search cost are common knowledge. Realized match values are not public information. Sellers do not observe them when bidding or pricing, and the buyer learns each product's match value only when she searches that product.

The game has three stages, summarized in Figure 1. Stage 1 is the market for prominence. Stages 2 and 3 form the product market.

Stage 1. Auction for prominence. Sellers simultaneously submit bids in a second-price auction. The higher bidder receives the top position and pays the losing bid, $b^{(2)}$. The other seller receives the second position and pays nothing. If bids are equal, each seller has probability one-half of receiving the top position. After the auction, subscripts identify positions rather than seller identities. Seller 1 is the top seller and seller 2 is the second seller.

Stage 2. Pricing. After learning their positions and the position payment, sellers simultaneously post prices $p_1, p_2 \geq 0$.

Stage 3. Search and purchase. The buyer observes both prices and searches the top seller's product for free, learning v_1. She may pay s to search the second seller's product and learn v_2. Before searching the second seller, she may buy from the top seller or exit. After searching the second seller, she may buy from either seller or exit. Returning to the top seller is free, and buying from the second seller requires search.

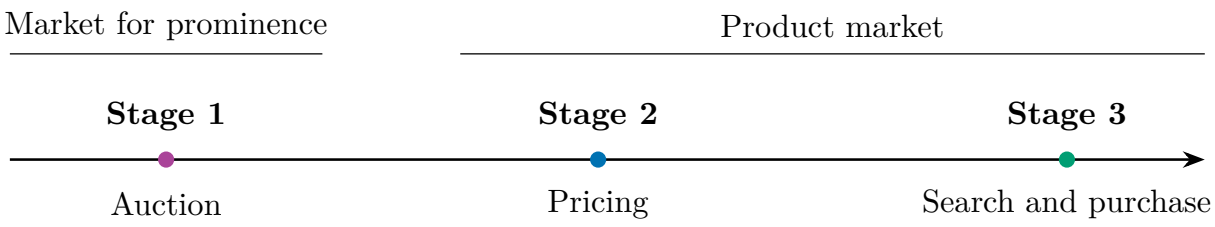


Figure 1: The three stages of the game

Let $x \in \{0,1\}$ indicate whether the buyer searches the second seller and let $a \in \{0,1,2\}$ denote her purchase decision, with $a = 0$ denoting exit. The buyer's payoff is

$$u(a,x) = \sum_{k=1}^{2} (v_k - p_k) \cdot \mathbf{1}_{\{a=k\}} - s \cdot x \tag{2}$$

Seller revenues, seller profits, and platform revenue are

$$\Pi_k = p_k \cdot \mathbf{1}_{\{a=k\}}, \qquad \pi_1 = \Pi_1 - b^{(2)}, \qquad \pi_2 = \Pi_2, \qquad \text{PR} = b^{(2)} \tag{3}$$

The position payment is sunk when sellers choose prices.

I solve the game backward. The buyer's search and purchase decisions determine each seller's demand at a given price pair. These demand functions yield a pricing equilibrium for the product market. The difference between the resulting expected revenues at the two positions determines the value of prominence and the sellers' auction bids.

2.2 Buyer search and seller demand

After observing (v_1, p_1, p_2), the buyer compares the expected gain from searching the second seller with its cost. With a bad top match, search offers a chance of finding a product worth buying. With a good top match, it offers a chance of buying a suitable product at a lower price. The following lemma specifies optimal choices, including choices at indifference.[3]

Lemma 1 (Optimal search cutoffs). *For $0 \le p_1, p_2 \le V$, the buyer's optimal search decision satisfies*

$$x^*(0, p_1, p_2) = 1 \quad \Longleftrightarrow \quad p_2 \le V - \frac{s}{q} \tag{4}$$

$$x^*(V, p_1, p_2) = 1 \quad \Longleftrightarrow \quad p_1 - p_2 > \frac{s}{q} \tag{5}$$

Thus, at a search tie the buyer searches after a bad top match and stops after a good top match. She buys an available product with the largest nonnegative purchase surplus or exits if all available products have negative purchase surplus. At ties, purchase is preferred to exit and the top seller is preferred to the second seller.

Proof. See Appendix A.1, which also specifies optimal behavior after prices outside $[0, V]$.□

Figure 2 summarizes the buyer's choices. The complete buyer decision tree is in Appendix A.1.

[3]The tie rules support the pricing equilibrium derived below. Appendix A.1 states them formally.

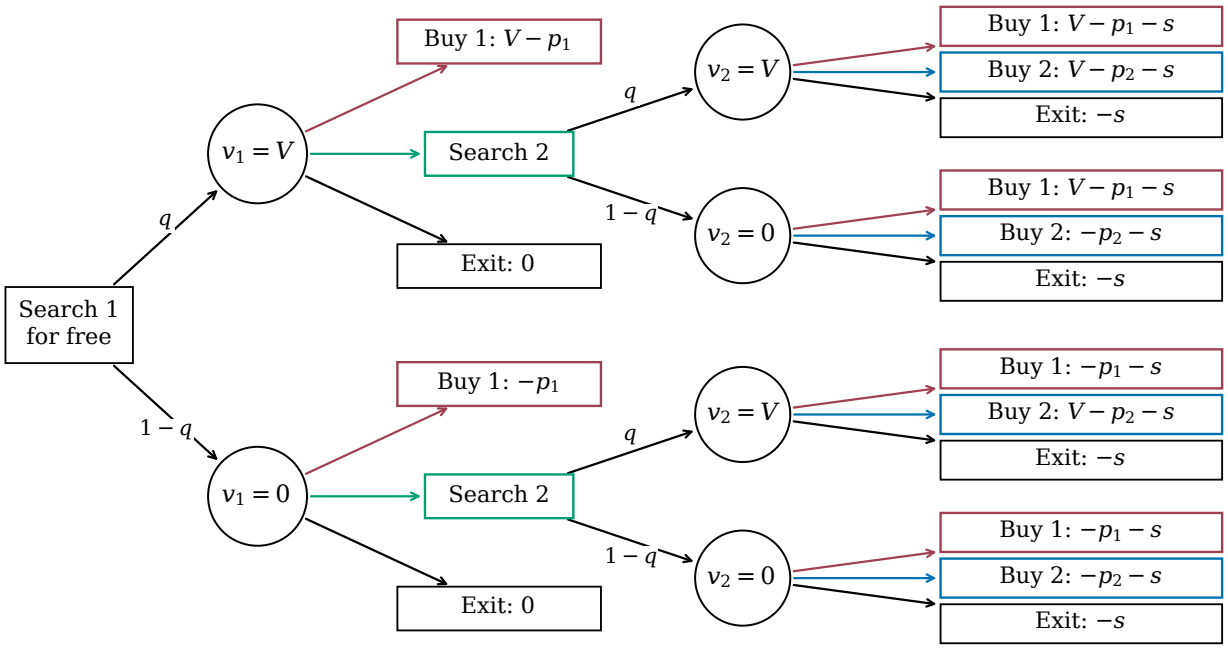


Figure 2: The buyer's search and purchase decisions

The buyer observes both prices and learns the top match for free. Search of the second seller costs s. Circles show realized match values and rectangles show choices.

For $0 < p_1 \leq V$ and $0 < p_2 \leq V - \frac{s}{q}$, let $D_k(p_1, p_2)$ be seller k's sale probability. The search cutoffs imply

$$\begin{array}{c|cc} & D_1(p_1,p_2) & D_2(p_1,p_2) \\ \hline p_1 - p_2 > \frac{s}{q} & q(1-q) & q \\ p_1 - p_2 \leq \frac{s}{q} & q & q(1-q) \end{array} \tag{6}$$

When a buyer with a good top match stops, the top seller serves all such buyers, while the second seller sells only when the top match is bad and the second match is good. When the price gap induces search, the second seller also attracts buyers with two good matches. The top seller then sells only when the top match is good and the second match is bad. Figures 3 and 4 show the purchase paths in the two cases. At a zero price, the purchase tie rule can change unit demand, but the zero-price seller's revenue is always zero.

2.3 Pricing equilibrium

Proposition 1 (No pure-strategy Nash equilibrium). *Given the buyer behavior in Lemma 1 and $0 < s < qV$, the pricing subgame has no pure-strategy Nash equilibrium.*

Proof. See Appendix A.2. □

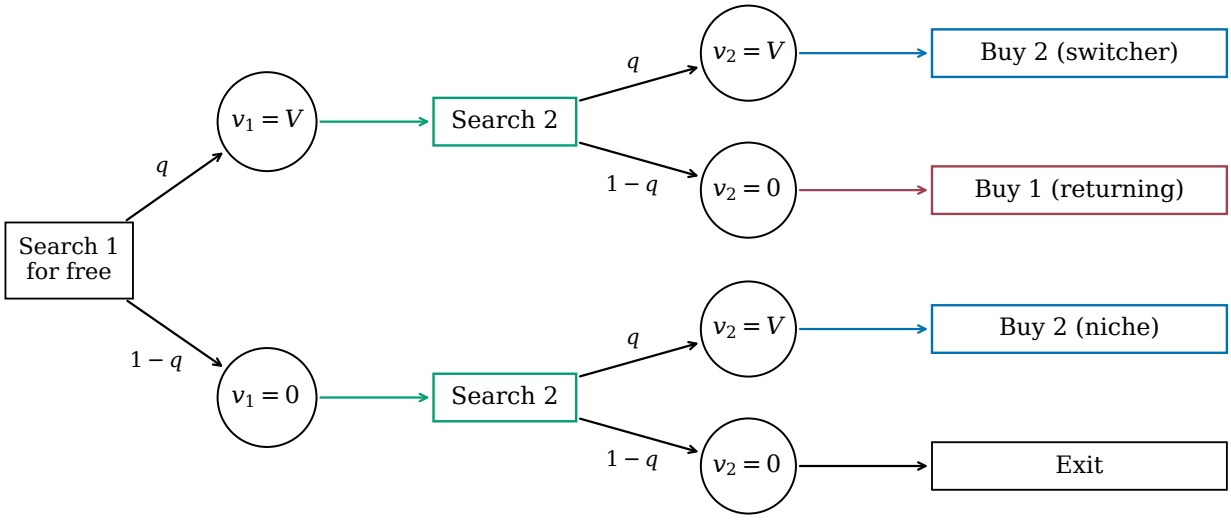


Figure 3: Search after a good top match when $p_1 - p_2 > \frac{s}{q}$

For $0 < p_1 \leq V$ and $0 < p_2 \leq V - \frac{s}{q}$, a buyer with a good top match searches and either buys from seller 2 after a good second match (switcher) or returns to seller 1 after a bad second match (returning). A buyer with a bad top match searches and buys from seller 2 after a good second match (niche), or exits after a bad second match.

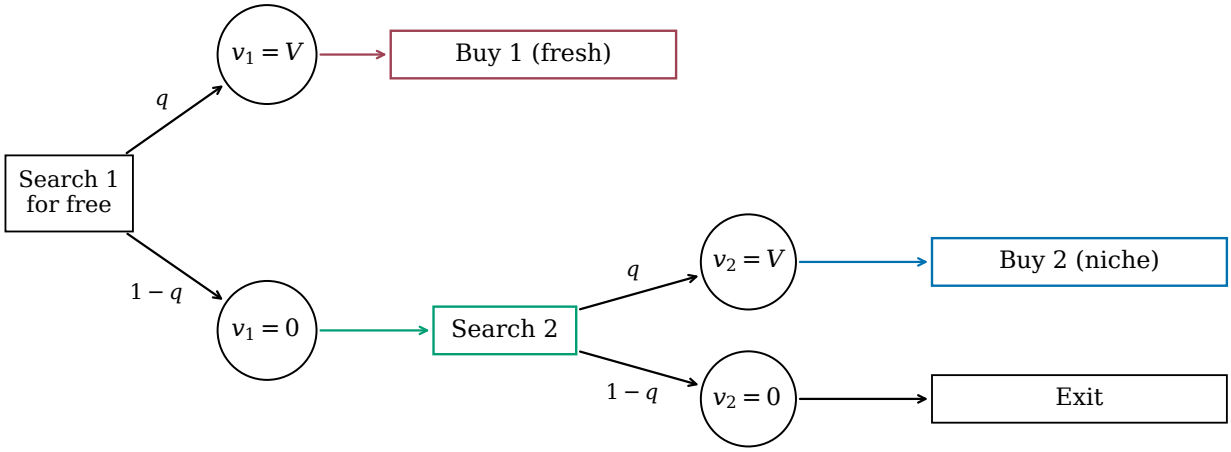


Figure 4: No search after a good top match when $p_1 - p_2 \leq \frac{s}{q}$

For $0 < p_1 \leq V$ and $0 < p_2 \leq V - \frac{s}{q}$, a buyer with a good top match buys from seller 1 without further search (fresh), leaving the second match unobserved. A buyer with a bad top match searches and buys from seller 2 after a good second match (niche), or exits after a bad second match.

On a constant-demand segment, each seller benefits from raising its price. At the search cutoff, the second seller can instead gain demand by slightly lowering its price. These opposing incentives motivate mixed pricing strategies. At deviation prices outside the domain in (6), demand follows the buyer's optimal strategy.

Definition 1 (Mixed-strategy pricing equilibrium). We seek distribution functions $F_1(p_1)$ and $F_2(p_2)$ with connected supports $[\underline{p}_1, \overline{p}_1]$ and $[\underline{p}_2, \overline{p}_2]$ such that

$$p_1^* \in \arg\max_{p_1 \in [0,\infty)} p_1 \mathbb{E}_{F_2}[D_1(p_1, p_2)], \qquad \forall p_1^* \in [\underline{p}_1, \overline{p}_1] \tag{7}$$

$$p_2^* \in \arg\max_{p_2 \in [0,\infty)} p_2 \mathbb{E}_{F_1}[D_2(p_1, p_2)], \qquad \forall p_2^* \in [\underline{p}_2, \overline{p}_2] \tag{8}$$

Theorem 1 (A mixed-strategy pricing equilibrium). *Given the buyer behavior in Lemma 1 and $0 < s < qV$, the following independent distributions constitute an equilibrium of the pricing subgame:*

$$F_1(p_1) = \frac{1}{q}\left[1 - \frac{(1-q)\left(V - \frac{s}{q}\right)}{p_1 - \frac{s}{q}}\right], \qquad \underline{p}_1 \le p_1 \le \overline{p}_1 \tag{9}$$

$$F_2(p_2) = \frac{1}{q}\left[1 - \frac{(1-q)V + s}{p_2 + \frac{s}{q}}\right], \qquad \underline{p}_2 \le p_2 < \overline{p}_2 \tag{10}$$

where

$$\begin{aligned} \underline{p}_1 &= (1-q)V + s, & \overline{p}_1 &= V \\ \underline{p}_2 &= (1-q)\left(V - \frac{s}{q}\right), & \overline{p}_2 &= V - \frac{s}{q} \end{aligned} \tag{11}$$

Both distributions are zero below their lower bounds and one at or above their upper bounds. The top seller's distribution is atomless. The second seller places mass $\frac{s}{qV}$ at $\overline{p}_2$.

Proof. Appendix A.3 verifies both optimality conditions against all nonnegative price deviations. □

The support bounds satisfy

$$\underline{p}_1 - \underline{p}_2 = \overline{p}_1 - \overline{p}_2 = \frac{s}{q} \tag{12}$$

A higher search cost raises the lower bound of the top seller's price distribution and lowers both bounds of the second seller's distribution. To induce search, the second seller must offer a larger price reduction relative to the top seller.

2.4 Bidding for prominence

Theorem 2 (Symmetric bidding equilibrium). *Suppose the sellers use the pricing equilibrium in Theorem 1 after every auction outcome. Their expected revenues before any position payment are*

$$\mathbb{E}[\Pi_1] = q[(1-q)V + s], \qquad \mathbb{E}[\Pi_2] = q(1-q)\left(V - \frac{s}{q}\right) \tag{13}$$

The value of prominence is their difference,

$$\mathbb{E}[\Pi_1] - \mathbb{E}[\Pi_2] = s \tag{14}$$

The symmetric bids $b_1 = b_2 = b^* = s$, together with these pricing strategies and optimal buyer behavior, constitute a subgame-perfect equilibrium. Platform revenue and expected seller profits are

$$\mathrm{PR} = s, \qquad \mathbb{E}[\pi_1] = \mathbb{E}[\pi_2] = (1-q)(qV - s) \tag{15}$$

Proof. See Appendix A.4. □

The auction payment absorbs the top seller's expected revenue advantage. A higher search cost raises the value of prominence and the equilibrium payment to the platform, while reducing both sellers' expected profits.

2.5 Welfare

The purchase decision is $a = 0$ for exit and $a = 1$ or $a = 2$ for purchase from the corresponding seller. The search indicator is $x = 1$ if the buyer searches the second seller and $x = 0$ otherwise. Consumer surplus is the buyer's payoff. Total surplus includes both sellers and the platform:

$$\mathrm{CS} = \sum_{k=1}^{2} (v_k - p_k) \cdot \mathbf{1}_{\{a=k\}} - s \cdot x \tag{16}$$

$$\mathrm{TS} = \mathrm{CS} + \pi_1 + \pi_2 + \mathrm{PR} = \sum_{k=1}^{2} v_k \cdot \mathbf{1}_{\{a=k\}} - s \cdot x \tag{17}$$

Prices and position payments are transfers. Search uses real resources, so total surplus depends on the product purchased and the search cost incurred. Search can increase surplus by revealing a valuable second product, but it can also incur a cost without changing the product ultimately purchased.

3 Experimental Design

3.1 Protocol

The experiment implements the three-stage market described above. Each market consists of two sellers and one buyer. Each product independently has a good match value of $V = 100$ experimental points with probability $q = 0.5$ and a bad match value of zero otherwise. I vary the cost of searching the second seller between sessions. The low-cost treatment has $s = 20$ and the high-cost treatment has $s = 40$. All other market rules are the same, and subjects participate in only one treatment.

Sellers bid for positions, learn their positions and the position payment, and then set prices. The buyer observes both prices and the top match before deciding whether to search the second seller. She can buy the second seller's product only after search and can return to the top seller for free. Bids and prices are restricted to integer values from 0 to 100. The continuous-price equilibrium provides the theoretical benchmark for this integer-grid implementation.

Roles are assigned at the beginning and remain fixed. In each round, the computer randomly and anonymously rematches subjects into groups of two sellers and one buyer and independently draws new match values. Subjects receive written instructions and answer three comprehension questions before completing two unpaid practice rounds and 30 incentivized rounds. At the end of each incentivized round, they receive feedback on the realized history of their market and their own payoff. Feedback concerns information revealed during play. The second match value is not disclosed when the buyer has not searched that product.

Every round begins with a balance of 200 experimental points, reset independently of earlier earnings. This balance covers possible losses from search and position payments. One of the 30 incentivized rounds is randomly selected for payment. The selected round's balance plus payoff is converted at 10 experimental points per dollar and added to a \$10 show-up fee. The welfare measures exclude the initial balance and show-up payment and measure surplus generated within the market. Appendix C reproduces the instructions, and Appendix D presents screenshots of the experimental interface.

3.2 Laboratory Implementation

I conducted six computerized sessions at the University of Virginia's Veconlab, with three sessions per treatment. There were 42 subjects in the low-cost treatment and 45 in the high-cost treatment. Five sessions had 15 subjects and one low-cost session had 12. The low-cost treatment included 28 sellers and 14 buyers, while the high-cost treatment included 30 sellers and 15 buyers. With 30 incentivized rounds per subject, the sample contains 420 low-cost and 450 high-cost market observations.

Each market in each round is one observation for the outcome comparisons, which treat these observations as independent. Search regressions use session-clustered standard errors.

3.3 Hypotheses

The hypotheses apply the continuous equilibrium to the experimental parameters. Numerical predictions average over equilibrium prices and match draws before their realization. Section 4.1 also compares observed outcomes with benchmarks conditional on the realized match draws. In both calculations, the buyer's decisions use only the information available at the time of choice. Appendix B gives the calculations.

Hypothesis 1 (Bidding and platform revenue). Raising the search cost from 20 to 40 raises equilibrium bids and platform revenue from 20 to 40.

The larger search disadvantage of the second seller increases the revenue advantage of the top position by 20. Competition for that position transfers the increase to the platform.

Hypothesis 2 (Prices). Top prices rise and second prices fall as search cost increases, with price distributions given by (9) and (10).

In the low-search-cost treatment, the top and second price supports are $[70, 100]$ and $[30, 60]$, with means of 81.59 and 49.93. In the high-search-cost treatment, the supports are $[90, 100]$ and $[10, 20]$, with means of 93.86 and 18.96. The top price distribution first-order stochastically dominates the second price distribution in each treatment.

Hypothesis 3 (Search). Search follows the match-specific cutoffs in Lemma 1 and becomes less frequent as search cost increases.

After a bad top match, the buyer searches when the second price is at most 60 in the low-search-cost treatment and at most 20 in the high-search-cost treatment. After a good top match, she searches when the price gap exceeds 40 and 80, respectively. At equilibrium prices, a buyer with a bad top match always searches. Search after a good top match occurs with probabilities 26.65% and 8.06%, so the overall search probabilities are 63.32% and 54.03%.

Hypothesis 4 (Revenues, profits, and welfare). Higher search cost raises the top seller's expected revenue and lowers the second seller's expected revenue, both sellers' expected profits, consumer surplus, and total surplus.

Expected top and second revenues change from 35 and 15 in the low-search-cost treatment to 45 and 5 in the high-search-cost treatment. Each seller's expected profit falls from 15 to 5, so combined seller profits fall from 30 to 10. Expected total surplus falls from 62.34 to 53.39 and expected consumer surplus falls from 12.34 to 3.39. These are unconditional

predictions. The realized-draw benchmarks used in the empirical comparison can differ because the observed frequencies of match pairs need not equal their ex ante probabilities.

4 Experimental Results

Consumers receive more surplus than equilibrium predicts even though total surplus is lower. Search cost affects surplus through both the expense of searching and the prices and decisions that determine whether search takes place. I first establish how the observed division of surplus differs from equilibrium, then examine the behavior behind those differences. Pricing and search determine what consumers buy and what sellers earn; bidding then determines how much of sellers' revenue is paid to the platform.[4]

4.1 Consumer surplus and the division of surplus

Consumers fare better than equilibrium predicts at both search costs. As Figure 5 and Table 1 show, consumer surplus averages 22.29 rather than 13.34 at the low search cost and 16.23 rather than 2.84 at the high search cost. It falls by 27% across treatments, compared with a 79% decline in the benchmark using the observed match draws. Higher search cost therefore reduces consumer surplus, but the observed loss is much less pronounced than the equilibrium comparison suggests. The behavioral question is which prices and search decisions account for this difference.

[4]The analysis uses all 30 incentivized rounds, with 420 low-cost and 450 high-cost markets. Treatment means give each market equal weight. Unless explicitly described as ex ante predictions, revenue and surplus benchmarks apply equilibrium strategies to the match draws observed in each treatment and average over equilibrium prices. Appendix B gives the calculations. Reported tests retain the market-level comparisons. Within-treatment tests use observed outcomes minus their corresponding benchmarks, and across-treatment comparisons use Welch tests. These tests treat market-rounds as independent and do not adjust for repeated participation within sessions.

Table 1: Welfare outcomes and equilibrium benchmarks

	$s = 20$ (Low)		$s = 40$ (High)		High minus low	
Outcome	Observed	Benchmark	Observed	Benchmark	Observed	Benchmark
Total surplus	58.62***	64.03	50.09***	54.23	-8.53	-9.80
Platform revenue	23.35***	20.00	23.83***	40.00	0.48	20.00
Consumer surplus	22.29***	13.34	16.23***	2.84	-6.06	-10.50
Combined seller profits	12.98***	30.69	10.03	11.39	-2.95	-19.29

Means are in experimental points per market over all 30 rounds ($N = 420$ in the low-search-cost treatment and $N = 450$ in the high-search-cost treatment). Stars test equality of observed means with the corresponding benchmark mean using paired market differences: $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$. Changes are high minus low. All tests are two-sided and treat markets as independent. Benchmarks use equilibrium strategies weighted by the observed match frequencies, averaging over equilibrium prices. They can differ from the ex ante predictions in Section 3.3. Seller profits subtract the payment to the platform.

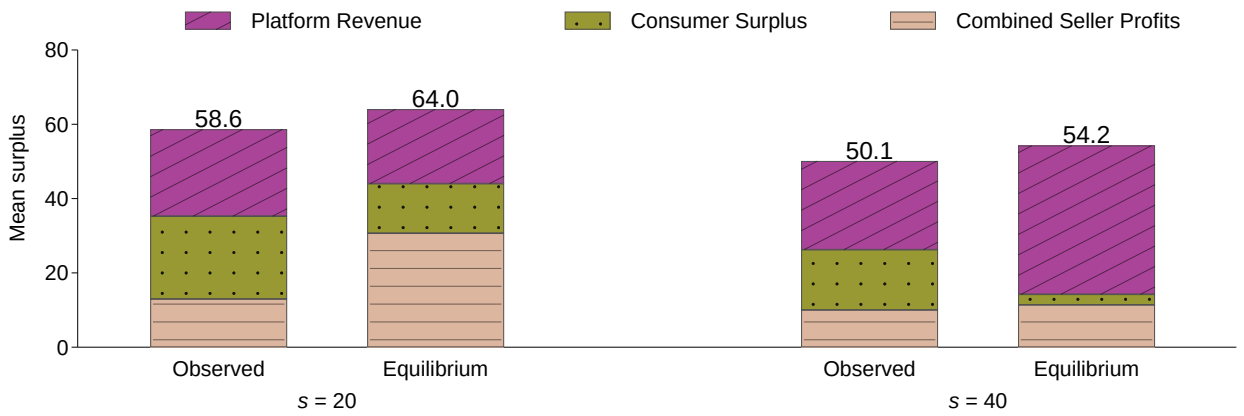


Figure 5: The level and distribution of surplus

Treatment means over all 30 rounds. Each bar divides total surplus into platform revenue, consumer surplus, and combined seller profits. Equilibrium benchmarks use the observed match draws and average over equilibrium prices. Values above the bars report total surplus.

The higher consumer surplus does not reflect a larger total surplus. Total surplus is below its benchmark in both treatments and falls from 58.62 to 50.09 as search cost rises. Consumers receive more of a smaller surplus, while combined seller revenue is lower than predicted. Prices help explain this difference because a lower payment benefits a buyer who purchases while reducing the seller's revenue from that transaction. Search decisions matter as well, because they determine whether the buyer incurs a further cost and discovers another product worth purchasing.

The decline in total surplus is not simply the higher cost charged for an unchanged amount of search. The frequency of search falls from 40.2% to 25.3%, so average search expenditure rises from 8.05 to 10.13. Meanwhile, the average match value of the product purchased falls from 66.67 to 60.22, counting zero when no product is purchased. Thus, both higher search expenditure and fewer purchases of good matches contribute to the decline. Relative to equilibrium, buyers spend less on search at each cost, but the forgone match value is larger than that saving. Explaining this pattern requires distinguishing buyers who stop because the available prices make search unattractive from buyers who stop despite a profitable opportunity to search.

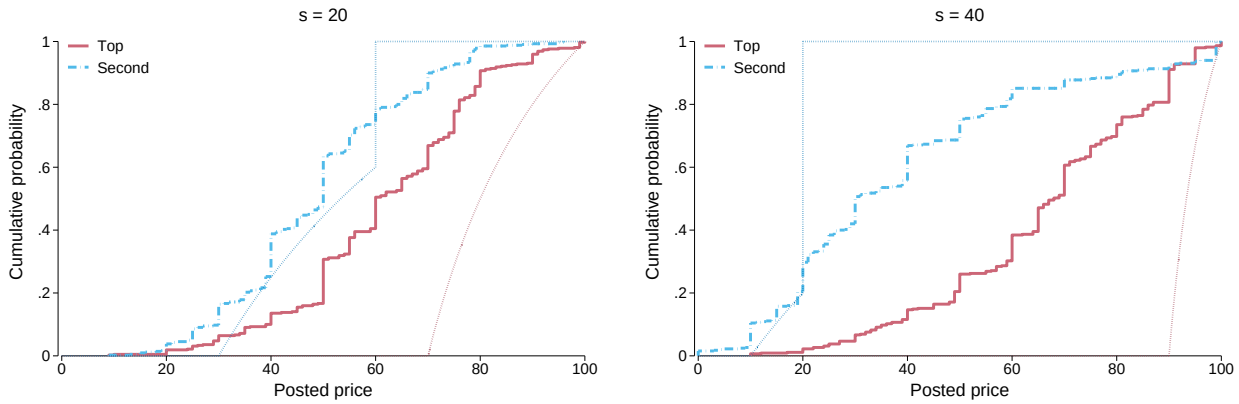
Result 1. Consumer surplus exceeds the benchmark in both treatments and falls by 27% as search cost rises, compared with a benchmark decline of 79%. Total surplus is lower than its benchmark at both costs.

Higher search cost does not generate the predicted increase in platform revenue. Platform revenue averages 23.35 and 23.83, despite the model's predicted increase from 20 to 40. Sellers retain 22.1% and 20.0% of total surplus after paying the platform. These shares have different implications across treatments. The low-cost share is far below its benchmark of 47.9%,

whereas the high-cost share is close to its benchmark of 21.0%. At the high cost, however, sellers earn less sales revenue and pay less to the platform than predicted. These departures largely offset each other, leaving profits close to their benchmark. I return to these payments after examining pricing and search.

4.2 Pricing and consumer search

Prices and the incentive to search. The price responses have the directions predicted by the model, but their magnitudes are substantially smaller. As search cost rises, the mean price at the prominent seller increases by 4.24 rather than the predicted 12.27, while the mean price at the non-prominent seller falls by 9.79 rather than 30.97. Figure 6 shows the resulting departures from the equilibrium distributions, and Table 2 reports the means.



(a) Low search cost

(b) High search cost

Figure 6: Observed and equilibrium price distributions

All 30 rounds. Red denotes the prominent seller and blue the non-prominent seller. Solid red and dash-dot blue curves are observed CDFs; dotted curves are equilibrium CDFs. Prices are those posted, whether or not a sale occurs.

Prices at prominent sellers are below equilibrium at both costs. Their means are 61.83 and 66.07, compared with 81.59 and 93.86. At non-prominent sellers, the low-cost mean is close to its benchmark, but the high-cost mean is 39.11 rather than 18.96. The price advantage of the non-prominent product is consequently much smaller than predicted, with a mean gap of 12.93 rather than 31.65 at the low cost and 26.96 rather than 74.90 at the high cost. Prices are also overdispersed relative to equilibrium at both positions.[5]

The prominent seller posts the higher of the two prices in 66.7% of low-cost markets and 81.6% of high-cost markets. These are comparisons within each market. Prominence is determined by the auction, before the sellers choose their prices.

[5]Observed standard deviations at the prominent position are 17.48 and 20.27, compared with 8.39 and 2.80 in equilibrium. At the non-prominent position, they are 16.30 and 25.36, compared with 10.57 and 2.44. These are descriptive comparisons.

Table 2: Bids and posted prices

	$s = 20$ (Low)		$s = 40$ (High)		High minus low	
Outcome	Observed	Benchmark	Observed	Benchmark	Observed	Benchmark
Prominent price	61.83***	81.59	66.07***	93.86	4.24	12.27
Non-prominent price	48.90	49.93	39.11***	18.96	-9.79	-30.97
Pooled bid	34.03***	20.00	38.79	40.00	4.76	20.00
Winning bid	44.71***	20.00	53.76***	40.00	9.05	20.00
Losing bid	23.35***	20.00	23.83***	40.00	0.48	20.00

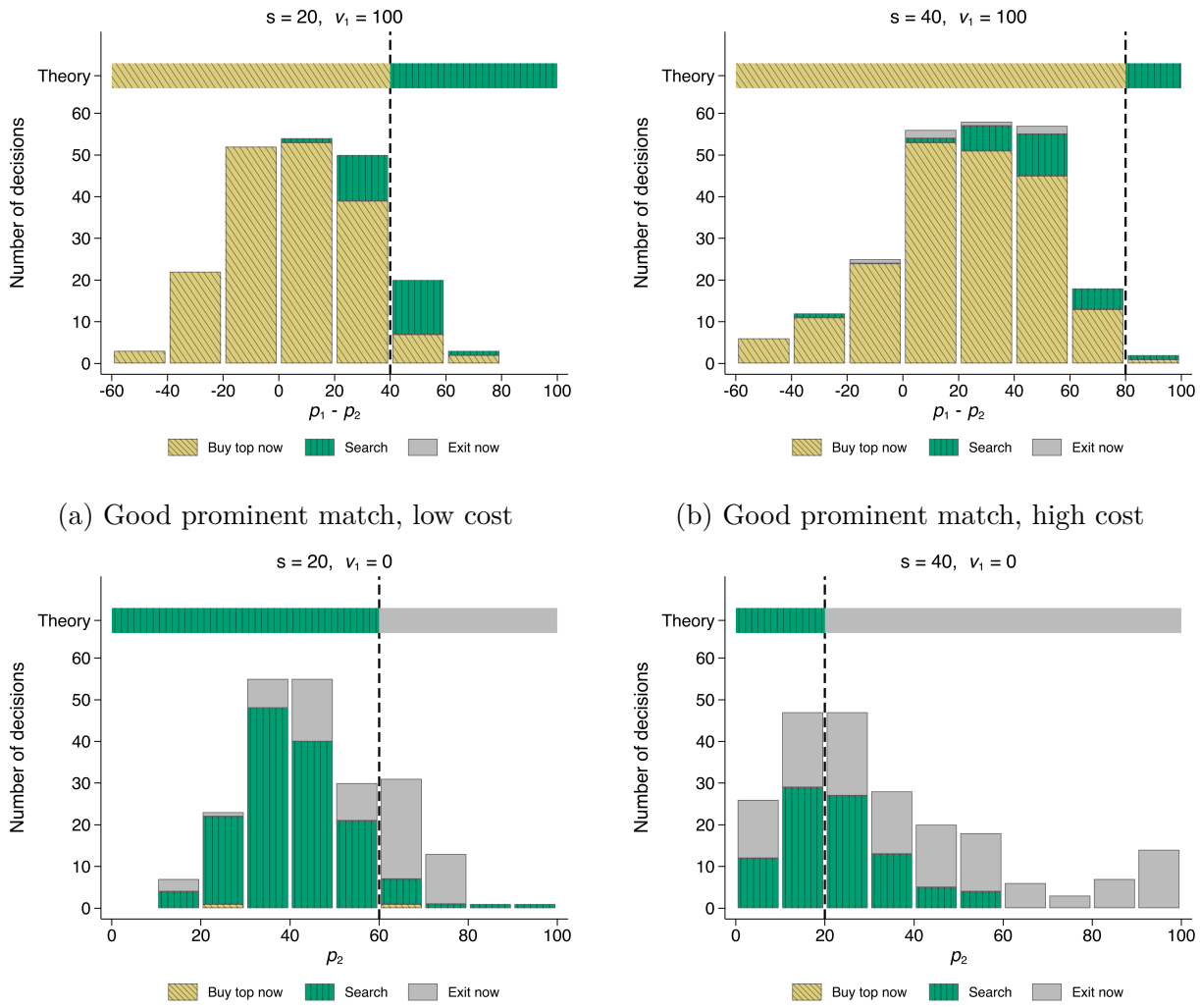
Means are in experimental points per market over all 30 rounds ($N = 420$ in the low-search-cost treatment and $N = 450$ in the high-search-cost treatment). Stars test equality of observed means with the corresponding benchmark mean using paired market differences: $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$. Changes are high minus low. All tests are two-sided and treat markets as independent. Benchmark prices are analytical equilibrium means and benchmark bids equal the search cost. The pooled bid is the mean of the two submitted bids in a market and contributes one observation. The losing bid equals the prominent seller's payment to the platform.

These price departures affect buyers differently depending on the match they find at the prominent seller. After a good match, a lower prominent price makes an immediate purchase more attractive, both by increasing the surplus from that purchase and by reducing the expected gain from further search. After a bad match, the price at the non-prominent seller determines whether inspecting its product is worthwhile. A price that is too high can deter search and prevent a sale even when that product would turn out to be a good match. This is especially relevant at the high search cost, where prices at non-prominent sellers are above their equilibrium benchmark.

Result 2. Prices respond to search cost in the predicted directions, but by less than predicted. Prices at prominent sellers are below equilibrium at both costs, while prices at non-prominent sellers are above equilibrium at the high cost. The resulting price gaps are smaller than predicted, and prices at both positions are overdispersed.

When buyers stop and when they search. Figure 7 follows the buyer's decision after observing the prominent product. After a good match, she can buy a product she already values or pay to investigate whether the other product offers a better deal. After a bad match, she must search to find a product with a positive match value. The same decision not to search can therefore represent acceptance of an attractive offer or a missed opportunity to find a suitable product.

After a good match, roughly 90% of decisions satisfy the optimal search rule in both treatments. The compressed price gap limits the occasions on which search is optimal, since search is strictly preferable only when $p_1 - p_2 > 2s$. At the high cost, just two of the 233 observations away from the cutoff satisfy this condition. Thus, the high compliance rate largely reflects buyers stopping when their inspected product already offers the better



(a) Good prominent match, low cost

(b) Good prominent match, high cost

(c) Bad prominent match, low cost

(d) Bad prominent match, high cost

Figure 7: Search decisions at observed prices

All 30 rounds. Bar heights show the frequency of observed prices or price differences; segments distinguish immediate purchase, search, and exit. The strip marked Theory gives the optimal action at each price. Cutoff ties appear in the figures but are excluded from strict-compliance rates.

Table 3: Search decisions relative to the optimal cutoff

Prominent match	Search cost	Strict cases	Compliant	Under-search	Over-search	Ties
Good	$s = 20$ (Low)	199	90.5%	9	10	5
Good	$s = 40$ (High)	233	89.7%	1	23	1
Bad	$s = 20$ (Low)	210	80.5%	32	9	6
Bad	$s = 40$ (High)	195	64.6%	20	49	21

Compliance classifies the search decision at observed prices. Exact search ties are excluded from the strict cases and reported separately. Under-search means stopping when search has strictly positive net expected value. Over-search means searching when stopping is strictly preferable. Counts of errors are not conditional error rates. Purchase optimality is a separate decision.

expected payoff. Lower prices at prominent sellers benefit these buyers directly and make further search less attractive.

After a bad match, search is strictly worthwhile when $p_2 < 100 - 2s$. Yet the non-prominent price lies above this cutoff in 46 of the 216 low-cost observations and 143 of the 216 high-cost observations with a bad prominent match. For these buyers, not searching is an optimal response to the prices they face. Their exit cannot be interpreted as under-search merely because another product remains uninspected. The high incidence of such prices at the high cost shows how sellers' pricing can restrict trade through its effect on the incentive to acquire product information.

Buyers also depart from the optimal rule at given prices, and these errors go in both directions. When search is strictly preferable after a bad match, 19.5% of low-cost buyers and 38.5% of high-cost buyers stop instead. When stopping is strictly preferable, 19.6% and 34.3% search anyway. Under-search can prevent a valuable purchase; over-search incurs a cost that exceeds the expected private gain from inspecting the other product.[6] Consequently, lower aggregate search does not mean that all buyers search too little. Buyers encounter different incentives, and some search when the observed prices make doing so unprofitable.

Purchase decisions are much closer to payoff maximization than search decisions. There are four purchase errors in the low-cost treatment and six in the high-cost treatment, counting choices both before and after further search. The principal difficulty is therefore acquiring information, rather than choosing among products whose matches are already known.

Search errors do not explain why consumer surplus exceeds equilibrium. At the observed prices, replacing actual search decisions with the optimal rule would raise expected consumer surplus by a further 1.02 at the low cost and 1.88 at the high cost.[7] The gains come from the prices buyers face, despite the losses from search errors. Lower prices at prominent sellers increase the surplus from immediate purchase after a good match. By contrast, high prices at non-prominent sellers can deter search after a bad match, and under-search can prevent purchases even when search is worthwhile. Thus, pricing benefits some consumers while pricing and search together leave other potentially valuable trades unrealized, reducing sales revenue as well as total surplus.

Result 3. Most buyers follow the optimal search rule after a good prominent match. After a bad match, some exit because the non-prominent price makes search unprofitable, while

[6]The under-search counts are 32 of 164 and 20 of 52; the over-search counts are 9 of 46 and 49 of 143. Table 3 reports the remaining counts and cutoff ties. The classification uses the information available to the buyer before search, not whether the uninspected product happens to be a good match.

[7]This calculation holds prices and the observed initial matches fixed, integrates the unobserved second match with $q = \frac{1}{2}$, and assumes optimal purchase choices after either search decision. Expected consumer surplus rises from 21.83 to 22.85 and from 16.08 to 17.96. These expected gains isolate search errors; they should not be added directly to realized consumer surplus, which also reflects the particular second matches and purchase choices. The calculation holds sellers' behavior fixed rather than predicting a new equilibrium with optimal buyers.

others under-search or over-search at the prices they face. At observed prices, search errors reduce expected consumer surplus. Purchase errors are rare.

4.3 Bidding, platform revenue, and seller profits

Pricing and search determine sellers' sales revenues, but the auction determines how much they retain. The model predicts that higher search cost increases the revenue advantage of prominence and therefore raises both bids and platform revenue. In the experiment, bids increase while platform revenue remains near 24 in both treatments.

Losing bids remain near 24 even as winning bids increase, explaining why higher submitted bids do not generate the predicted increase in platform revenue (Figure 8). The mean pooled bid rises from 34.03 to 38.79, while the mean winning bid rises from 44.71 to 53.76. The mean losing bid, however, changes from 23.35 to 23.83. Because the auction charges the losing bid, the platform does not receive the higher amount submitted by the winner. Moreover, the high-cost pooled mean being close to the equilibrium bid of 40 masks a substantial separation between winning and losing bids. It does not imply that sellers bid close to 40 individually.

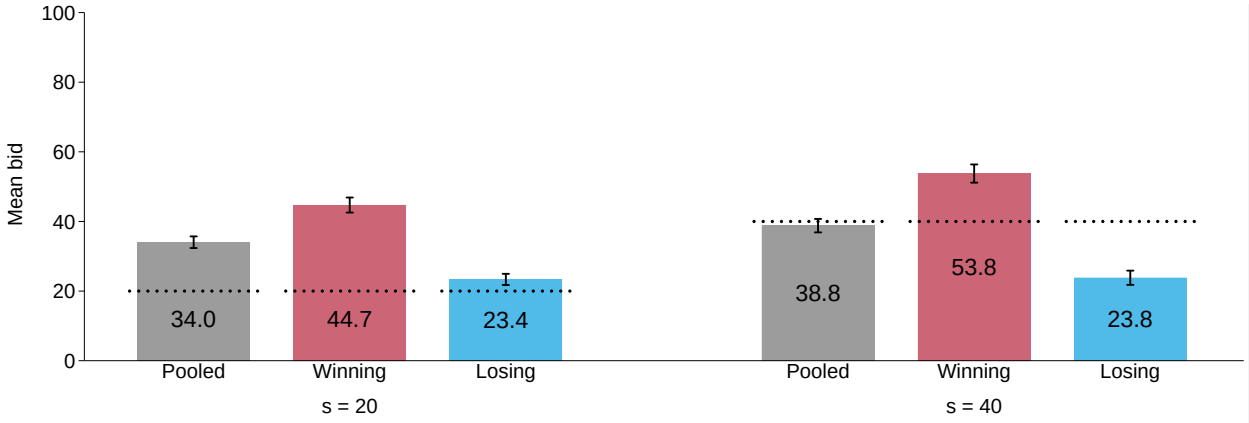


Figure 8: Pooled, winning, and losing bids

Means over all 30 rounds. The pooled bid averages both bids within a market. The losing bid is the payment to the platform. Error bars are 95% confidence intervals treating markets as independent. Horizontal lines indicate equilibrium bids.

Result 4. The mean pooled bid rises by 4.76 as search cost increases, compared with the predicted increase of 20. Platform revenue, determined by the losing bid, rises by 0.48 rather than 20.

These auction payments explain why seller profits can be near their benchmark even when sales revenues are far below it. Table 4 shows that both positions earn less sales revenue than predicted. At the low cost, combined revenue is 14.36 below its benchmark and the platform receives 3.35 more than predicted. Both departures reduce sellers' combined profit. At the

Table 4: Revenues and profits by position

	$s = 20$ (Low)		$s = 40$ (High)		High minus low	
Outcome	Observed	Benchmark	Observed	Benchmark	Observed	Benchmark
Prominent revenue, Π_1	27.67***	34.17	30.89***	46.73	3.22	12.57
Non-prominent revenue, Π_2	8.66***	16.52	2.97***	4.66	-5.69	-11.86
Combined revenue	36.33***	50.69	33.86***	51.39	-2.47	0.71
Revenue advantage, $\Pi_1 - \Pi_2$	19.01	17.64	27.91***	42.07	8.91	24.43
Platform revenue	23.35***	20.00	23.83***	40.00	0.48	20.00
Prominent profit, π_1	4.32***	14.17	7.06	6.73	2.74	-7.43
Non-prominent profit, π_2	8.66***	16.52	2.97***	4.66	-5.69	-11.86
Profit difference, $\pi_1 - \pi_2$	-4.34	-2.36	4.09	2.07	8.43	4.43

Means are in experimental points per market over all 30 rounds ($N = 420$ in the low-search-cost treatment and $N = 450$ in the high-search-cost treatment). Stars test equality of observed means with the corresponding benchmark mean using paired market differences: $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$. Changes are high minus low. All tests are two-sided and treat markets as independent. Benchmarks weight equilibrium outcomes by observed match frequencies. The model's ex ante profit difference is zero at both costs; nonzero benchmark differences here reflect the realized draws. Revenue excludes the prominence payment, which is subtracted from prominent profit. Positions are assigned by the auction.

high cost, combined revenue is 17.53 below its benchmark, but the platform receives 16.17 less than predicted. The lower payment offsets most of the revenue shortfall, leaving combined profit close to its benchmark. This is the reason the seller share is near its predicted level at the high cost; it is not evidence that pricing or auction behavior is close to equilibrium.

The cost of prominence also changes the comparison between the two positions. At the low cost, the prominent seller's observed revenue advantage is 19.01, less than the mean payment of 23.35. Its mean profit is consequently lower than the non-prominent seller's. At the high cost, the revenue advantage is 27.91 and the mean payment is 23.83, leaving the prominent seller with higher mean profit. Occupying the prominent position generates more sales revenue in both treatments, but its payment determines whether that advantage survives in net profit.

The model's ex ante prediction remains equal profits at the two positions, with each seller earning 15 at the low cost and 5 at the high cost. The nonzero benchmark profit differences in Table 4 arise because the table weights equilibrium outcomes by the match frequencies observed in each treatment. They reflect those realized draws, not a theoretical prediction that one position has higher expected profit.[8]

Mean submitted bids exceed the observed revenue advantage in both treatments, but this comparison alone does not establish overbidding relative to sellers' actual incentives. Bids are chosen before prices and matches are realized, and the sellers assigned to the two positions need not have identical pricing behavior. The observed position averages show how

[8]The four match pairs each have probability $\frac{1}{4}$ in the model. Weighting them equally gives a profit difference of zero at both costs. Using the observed frequencies instead gives -2.36 and 2.07. The difference between these adjusted benchmarks is likewise sample-dependent. Appendix B reports both calculations.

revenue and payments divide the proceeds from trade; evaluating whether a seller bids or prices optimally against the other participants requires its expected payoff from alternative choices.

Result 5. Sellers retain 20–22% of total surplus after paying for prominence. At the low search cost, lower sales revenue and an above-equilibrium auction payment jointly reduce their profits. At the high cost, a below-equilibrium payment offsets most of the revenue shortfall.

The welfare departures therefore have distinct sources. Prices at prominent sellers below equilibrium benefit consumers and reduce sellers' revenue per sale, while missed trades further reduce sales revenue. Search errors diminish consumers' gains at the prices they face. The losing bid then divides sellers' revenue between platform revenue and seller profits. Its near-constant level explains the weak response of platform revenue, while its departures from the equilibrium payment deepen the seller-profit shortfall at the low cost and offset much of it at the high cost.

5 Empirical Search Rules

Search varies with prices in the directions predicted by the model. After a bad match with the prominent product, search is primarily associated with the non-prominent price. After a good match, it rises with the prominent price and falls with the non-prominent price. Buyers therefore respond to search incentives without consistently following the optimal cutoff. I examine these relationships using separate regressions for the two search-cost treatments, allowing the price associations to depend on the match already observed.

Let $G = 1$ denote a good match with the prominent product and $G = 0$ a bad match, and let r denote the round. For each search cost, I estimate the following logit model using all 30 incentivized rounds.

$$\Pr(x = 1 \mid G, p_1, p_2, r, s) = \Lambda(\eta_s)$$
$$\eta_s = \alpha_s + \gamma_s G + \beta_{1s} p_1 + \beta_{2s} p_2 + \delta_{1s} G p_1 + \delta_{2s} G p_2 + \rho_s r \qquad (18)$$

Here Λ is the logistic distribution function. The interactions $G \times p_1$ and $G \times p_2$ allow the price associations to differ between match states. All coefficients can differ by treatment, while the round trend is common to the two match states within each treatment. The estimates

describe conditional associations at the prices buyers observe, rather than causal effects of experimentally assigned prices.[9]

The theory gives a different role to prices in the two match states. After a bad match, the prominent product offers no positive surplus, so its price should not affect whether inspecting the other product is worthwhile. After a good match, the buyer compares the expected gain from searching with the surplus already available. Only the price difference should matter, so raising both prices equally leaves the incentive to search unchanged. These predictions imply the following restrictions on the logit index.

$$\begin{aligned} H_B: &\quad \beta_{1s} = 0 \\ H_G: &\quad \beta_{1s} + \delta_{1s} + \beta_{2s} + \delta_{2s} = 0 \end{aligned} \tag{19}$$

The first restriction sets the prominent-price coefficient to zero after a bad match. The second requires equal and opposite price coefficients after a good match. Neither imposes a deterministic search cutoff on the fitted probabilities.

Table 5 reports the coefficient estimates for the 420 low-cost and 450 high-cost decisions, with standard errors clustered by session.[10]

After a bad match, search is primarily associated with the price of the product the buyer has not inspected. A higher non-prominent price is associated with less search in both treatments, whereas the association with the prominent price is much smaller. The average marginal effects for a 10-unit increase in the non-prominent price are -12.57 and -9.06 percentage points at the low and high search costs, compared with 1.99 and 2.08 for the prominent price.[11] The tests do not reject H_B, with nominal p-values of 0.2696 and 0.2390. Thus, the under-search and over-search documented in Section 4.2 coexist with a decline in search as the uninspected product becomes less attractive, consistent with the model.

After a good match, both prices matter in the predicted directions. Search is more frequent when the prominent product is expensive and less frequent when the non-prominent product is expensive, consistent with buyers comparing the offer they already know with the expected gain from another inspection. The two price associations are not equally strong, however. The average marginal effects for prominent and non-prominent prices are 5.56 and -9.23 at the low search cost, compared with 3.09 and -3.54 at the high cost. An equal increase in both prices therefore has an implied average derivative of -3.67 percentage points

[9] The model includes only information available when the buyer decides whether to search. It excludes the hidden second match and subsequent purchase outcome. After a bad match, the two price coefficients in the index are β_{1s} and β_{2s}; after a good match, they are $\beta_{1s} + \delta_{1s}$ and $\beta_{2s} + \delta_{2s}$.

[10] There are only three sessions in each treatment, so nominal session-clustered Wald p-values are sensitive to the variance estimate and should be interpreted cautiously.

[11] These are scaled local derivatives, not exact probability changes from a finite price increase. For each treatment and match state, I evaluate the derivatives at all observed price and round vectors, set G to the indicated state, and average. Price units are experimental points.

Table 5: Logit estimates of search decisions

	$s = 20$ (Low)	$s = 40$ (High)
Top price, p_1	0.0113	0.0105
	(0.0103)	(0.0089)
Second price, p_2	-0.0716	-0.0458
	(0.0042)	(0.0067)
Good top match, G	-5.6709	-3.8153
	(0.4016)	(0.6062)
$G \times p_1$	0.0689	0.0248
	(0.0047)	(0.0080)
$G \times p_2$	-0.0616	0.0054
	(0.0077)	(0.0092)
Round	-0.0116	-0.0281
	(0.0103)	(0.0350)
Constant	3.7778	0.9655
	(1.3355)	(0.5842)
Observations	420	450
Sessions	3	3

The dependent variable equals one if the buyer searches the second seller's product. Each column is a separate treatment regression using all 30 incentivized rounds. Parentheses contain standard errors clustered by session. Prices enter in experimental points. The entries are logit coefficients, not changes in search probability. G equals one for a good top match and zero for a bad top match.

per 10 units at the low cost and -0.45 at the high cost. The high-cost estimate is much closer to the model's prediction that only the price gap matters.

The estimated search rules retain the predicted directions of the price effects but depart from the restriction that only the price gap matters after a good match. The nominal tests reject the equal-and-opposite restriction H_G in both treatments ($p = 0.0015$ and $p = 5.63 \times 10^{-6}$), subject to the small number of session clusters noted above. The estimated relationships help interpret the earlier welfare findings. Lower prominent prices are associated with less search after a good match, while high non-prominent prices are associated with less search after a bad match. The former makes an immediate purchase more attractive; the latter can discourage the inspection needed for a sale. Whether sellers choose the revenue-maximizing prices against these observed responses is a separate question that these regressions do not test.

6 Agent Quantal Response Equilibrium

The preceding sections show that sellers adjust their bids and prices in the directions predicted by the model, but much less strongly. Why does higher search cost produce such a limited response? In equilibrium, the predicted increase in bids reflects the additional revenue that

prominence generates when sellers price and buyers search optimally. Yet the observed pricing and search decisions depart from these strategies. If sellers anticipate such departures, their expected revenue advantage from prominence could increase less than predicted, giving them less reason to raise their bids. I use the agent quantal response equilibrium (AQRE) of McKelvey and Palfrey (1998) to examine whether noisy responses throughout the game can jointly account for the compressed price gap and the observed increase in bids.

6.1 Specification and estimation

In the AQRE, an agent at each information set chooses higher-payoff actions more often, without always choosing a best response. Agents belonging to the same player share that player's payoff function, but their errors are independent across information sets. Each agent anticipates the noisy decisions of other players and its own player's later agents. Thus, a buyer considering search anticipates her subsequent purchase choices, and a seller bidding for prominence anticipates noisy pricing as well as noisy buyer decisions. In equilibrium, anticipated and actual choice probabilities coincide.

I use the logit specification with one common precision parameter $\lambda \geq 0$ across players, decision stages, and treatments. At $\lambda = 0$, choices are uniform over the available menu; a larger λ makes choices more sensitive to expected payoff differences. Bids and prices use the experimental menu $\{0, 1, \ldots, 100\}$. The buyer's menus retain all available purchases, search, and exit, including actions that are dominated in monetary payoff. The four components below are solved backward, evaluating earlier payoffs under subsequent quantal responses as described by Goeree et al. (2020, Section VII).

Choice after search. The buyer chooses $c \in \{\text{buy } 1, \text{buy } 2, \text{exit}\}$. Write $U(c \mid p_1, p_2, v_1, v_2)$ for the corresponding payoff before the sunk search cost, equal to $v_1 - p_1$, $v_2 - p_2$, or zero, respectively. The behavioral strategy is

$$\sigma(c \mid p_1, p_2, v_1, v_2) = \frac{\exp\{\lambda U(c \mid p_1, p_2, v_1, v_2)\}}{e^{\lambda(v_1 - p_1)} + e^{\lambda(v_2 - p_2)} + 1} \tag{20}$$

Subtracting the sunk search cost from every payoff leaves these probabilities unchanged.

Choice before search. The buyer initially chooses $a \in \{\text{buy } 1, \text{search}, \text{exit}\}$, knowing v_1 but not v_2. The payoffs from immediate purchase and exit are $v_1 - p_1$ and zero. Her expected payoff from search incorporates her own later noisy purchase decision and is given by

$$\begin{aligned} U(\text{search} \mid p_1, p_2, v_1) = -s + \mathbb{E}_{v_2}\Big[&\sigma(\text{buy } 1 \mid p_1, p_2, v_1, v_2)(v_1 - p_1) \\ &+ \sigma(\text{buy } 2 \mid p_1, p_2, v_1, v_2)(v_2 - p_2)\Big] \end{aligned}$$

The expectation uses the independent match distribution with $q = \frac{1}{2}$. The initial choice probabilities are

$$\sigma(a \mid p_1, p_2, v_1) = \frac{\exp\{\lambda U(a \mid p_1, p_2, v_1)\}}{e^{\lambda(v_1 - p_1)} + e^{\lambda U(\text{search} \mid p_1, p_2, v_1)} + 1} \tag{21}$$

Thus, the value of search uses the monetary payoffs generated by subsequent quantal responses, rather than assuming an optimal purchase afterward.

Pricing. Let $D_k(p_1, p_2)$ be the sale probability at position $k \in \{1, 2\}$ implied by these buyer strategies, averaging over the match draws. The pricing functions jointly satisfy the following equations for every $p_1, p_2 \in \{0, 1, \ldots, 100\}$.

$$\begin{aligned} f_1(p_1) &= \frac{\exp\left\{\lambda p_1 \sum_{p_2} D_1(p_1, p_2) f_2(p_2)\right\}}{\sum_{p_1'} \exp\left\{\lambda p_1' \sum_{p_2} D_1(p_1', p_2) f_2(p_2)\right\}} \\ f_2(p_2) &= \frac{\exp\left\{\lambda p_2 \sum_{p_1} D_2(p_1, p_2) f_1(p_1)\right\}}{\sum_{p_2'} \exp\left\{\lambda p_2' \sum_{p_1} D_2(p_1, p_2') f_1(p_1)\right\}} \end{aligned} \tag{22}$$

Each pricing function must be a logit response to the other, with demand incorporating noisy search and purchase. The auction payment is sunk and cancels from the comparison of pricing payoffs.

Bidding. Expected product-market revenues and the value of prominence are

$$\Pi_k = \sum_{p_1, p_2} p_k D_k(p_1, p_2) f_1(p_1) f_2(p_2), \qquad \Delta\Pi = \Pi_1 - \Pi_2$$

Given the opponent's bidding function g, a seller's expected profit from bid b is

$$\pi(b; g) = \Pi_2 + \sum_{b'} g(b')(\Delta\Pi - b') \left[\mathbf{1}\{b > b'\} + \frac{1}{2}\mathbf{1}\{b = b'\}\right]$$

The symmetric bidding function solves

$$g(b) = \frac{\exp\{\lambda \pi(b; g)\}}{\sum_{b'} \exp\{\lambda \pi(b'; g)\}}, \qquad b \in \{0, 1, \ldots, 100\} \tag{23}$$

Here the losing bid determines the payment, and equal bids receive an equal chance of winning. Symmetry means identical bidding functions, not equal realized bids. The prominence value entering this fixed point incorporates both noisy prices and noisy buyer decisions; it is not imposed to equal s.[12]

Estimated precision. I estimate one precision parameter jointly from bids, prices, and buyer choices in both treatments. For each candidate precision λ, I solve the full AQRE in each treatment and evaluate the probabilities it assigns to these observed decisions. Buyer-choice probabilities are evaluated at the prices and matches known at the corresponding information set. I then sum the log probabilities across all reached decisions and numerically maximize this joint log likelihood over $\lambda \geq 0$. The same candidate precision governs both treatments and all four decision layers; the estimate is not obtained by fitting each stage separately.

The data favor responses to payoff differences over uniform randomization. Table 6 reports $\widehat{\lambda} = 0.1146$, with a session-clustered standard error of 0.0101.[13] The uniform-choice null, $\lambda = 0$, is rejected ($p < 0.001$).[14] The fitted bid and price distributions then show how these responses affect competition for prominence.

Table 6: Estimated AQRE precision

	Estimate	Standard error
Common precision, $\widehat{\lambda}$	0.1146	0.0101

Joint maximum likelihood using all 870 markets. The standard error uses the observed likelihood curvature and score contributions aggregated over six sessions, with the cluster variance multiplied by $\frac{6}{5}$. Inference with six sessions is approximate. Payoffs use the experimental scale; precision is common across both treatments and all decisions.

6.2 Fitted bids, prices, and buyer choices

The AQRE reproduces the small increase in pooled bids. Their fitted means are 33.51 and 38.42, an increase of 4.90, close to the observed increase of 4.76. Figure 9 compares the

[12]The calculation selects the position-specific pricing and symmetric bidding solutions connected to uniform play as precision increases, using the same pricing continuation after each auction history. This is a maintained equilibrium selection, not a claim of global uniqueness.

[13]Payoffs are measured in the experimental units used in the decision screens. Expressing the same payoffs in dollars instead gives $\widehat{\lambda} = 1.1457$ and identical choice probabilities. The likelihood includes 1,740 bids, 1,740 prices, 870 initial buyer choices, and 283 choices after search.

[14]The maximized log likelihood is −15,949.61, compared with −17,327.32 under uniform choice, giving a likelihood-ratio statistic of 2,755.42. Because zero is the boundary of $\lambda \geq 0$, the one-sided asymptotic reference is an equal mixture of a point mass at zero and χ^2_1. This test uses the maintained likelihood assumptions. An approximate one-sided t_5 comparison using the session-clustered standard error also gives $p < 0.001$. Fits omitting one session at a time outperform uniform choice in every held-out session.

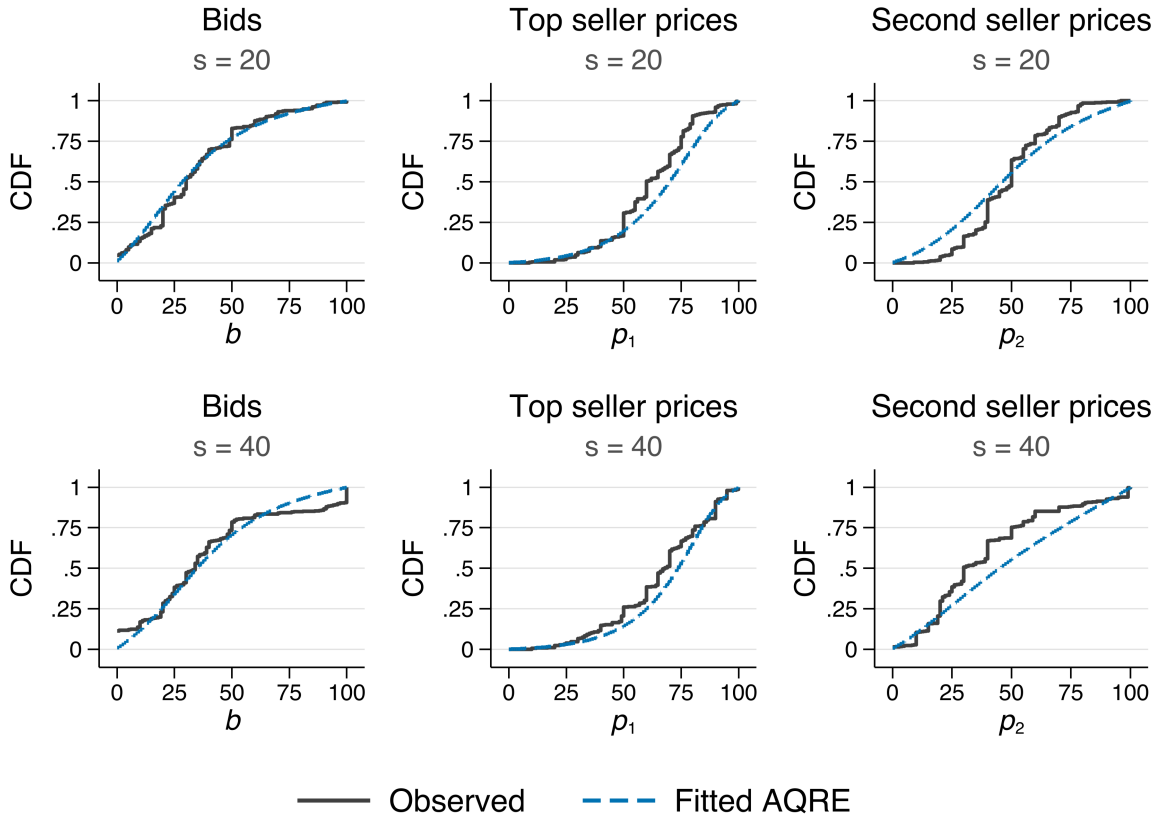


Figure 9: Observed and fitted bid and price distributions

All fitted distributions use the common estimate $\widehat{\lambda} = 0.1146$. Rows distinguish search costs. Top and second seller refer to the prominent and non-prominent positions.

full distributions. The value of prominence under the fitted continuation is 17.62 at the low search cost and 27.62 at the high cost, so the auction agents face a smaller increase in the revenue advantage than under the fully rational benchmark. The fitted losing bid rises from 20.17 to 24.71. This response is much smaller than the equilibrium increase of 20, although it remains larger than the observed increase of 0.48.

The AQRE reproduces the compressed price gap, with prices at prominent sellers remaining higher on average than at non-prominent sellers but below the fully rational benchmark. Their fitted means rise from 66.71 to 70.19, compared with 61.83 to 66.07 in the data. The fitted price gaps, 19.22 and 22.94, are consequently much smaller than the equilibrium gaps of 31.65 and 74.90. The fitted non-prominent means are 47.49 and 47.25, compared with 48.90 and 39.11 in the data.

Noisy buyer decisions determine sellers' demand at each pair of prices. Figures 10 and 11 show these decisions at observed and fitted prices. Bar heights give the frequency of each price bin, and colored segments distinguish purchase, search, and exit.[15]

The fitted AQRE captures the observed increase in pooled bids and the compressed price gap using one common precision parameter. Noise affects bidding both through sellers' responses to expected profits and through the pricing and buyer decisions that determine those profits.

[15]Expected counts are scaled to the observed number of buyers in each treatment and prominent-match group. Both prices enter the underlying AQRE buyer choice probabilities. The plots aggregate over price pairs within each price-gap bin after a good match and over prominent prices within each non-prominent-price bin after a bad match.

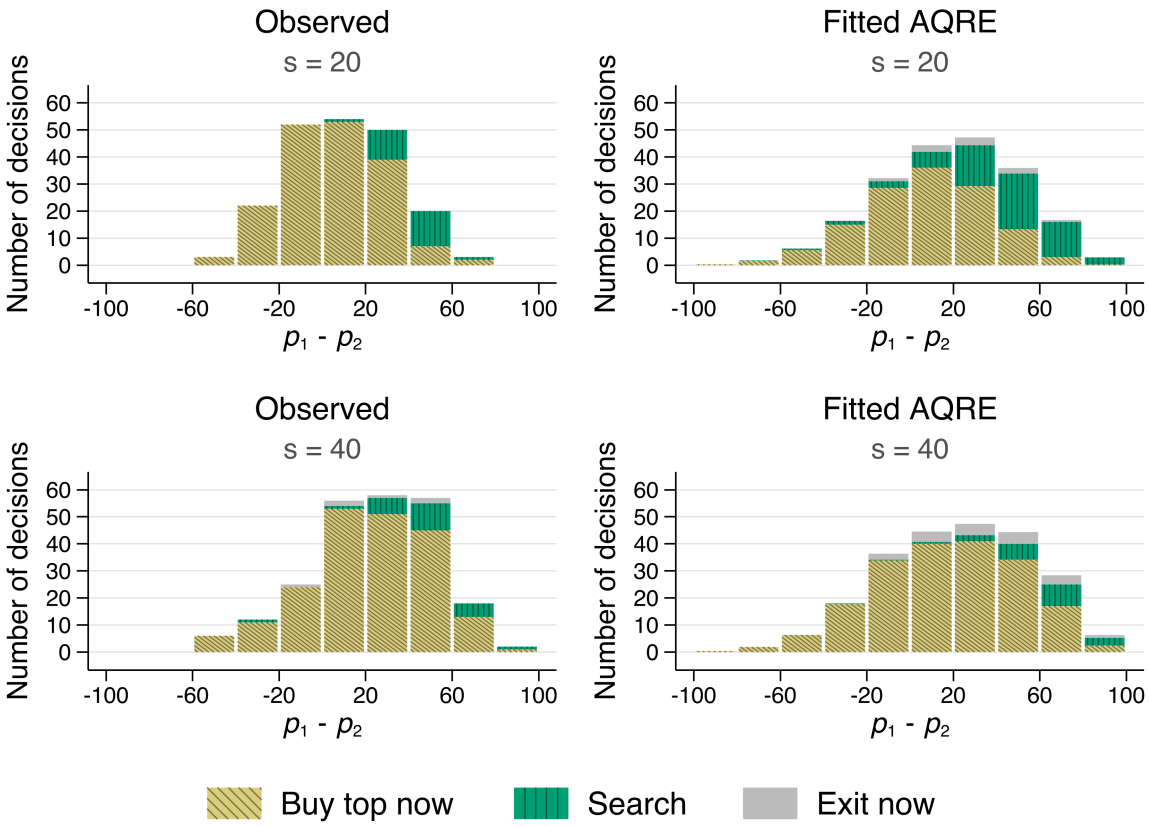


Figure 10: Observed and fitted buyer choices after a good prominent match

Rows show $s = 20$ and $s = 40$; columns show observed counts and AQRE expected counts. Bar heights give the number of decisions in each price-gap bin, with segments for immediate purchase, search, and exit. The AQRE generates both prices and choices. Cutoff ties are included.

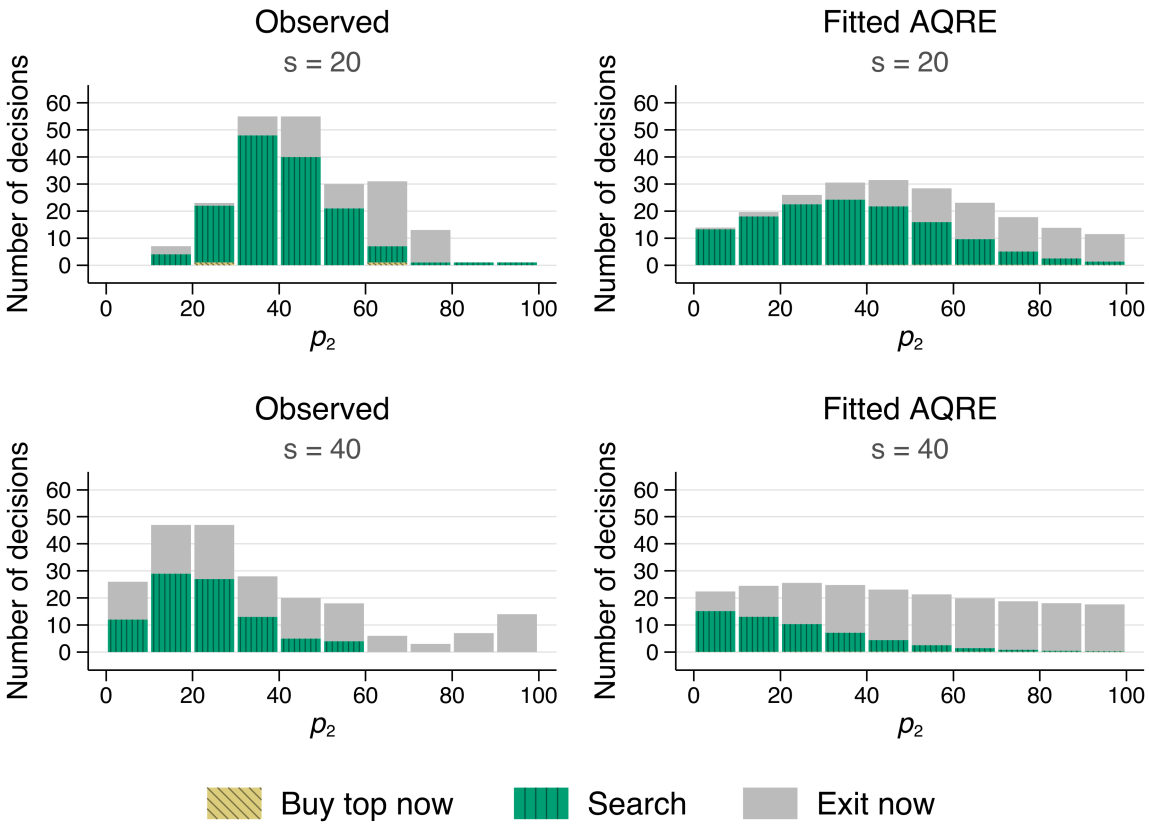


Figure 11: Observed and fitted buyer choices after a bad prominent match

Rows show $s = 20$ and $s = 40$; columns show observed counts and AQRE expected counts. Bar heights give the number of decisions in each non-prominent-price bin, with segments for immediate purchase, search, and exit. The AQRE generates both prices and choices. Cutoff ties are included.

7 Conclusion

This paper studies a market in which sellers compete for prominence before setting prices and consumers decide how far to search. The value of the prominent position depends on the pricing and search decisions that follow its allocation. A laboratory experiment varies the cost of searching the second seller's product and observes how surplus is divided among consumers, sellers, and the platform.

Consumer surplus exceeds its equilibrium benchmark at both search costs and falls by 27% as search cost rises, compared with 79% in the benchmark using observed matches. Prices at prominent sellers below equilibrium benefit consumers, while search errors reduce their expected surplus at the prices they face. Sellers earn less revenue per sale at those lower prices, and pricing and search decisions also leave some potentially valuable trades unrealized. Losing bids determine the subsequent division of sales revenue: platform revenue remains near 24 in both treatments, and sellers retain 20–22% of total surplus. At the high search cost, a payment below equilibrium offsets most of the sales-revenue shortfall, explaining why seller profits are closer to their benchmark there.

The AQRE captures the observed increase in pooled bids and the compressed price gap with one common precision parameter. Its positive estimate indicates that choices respond to expected payoff differences, although less sharply than under best responses. In this account, sellers value prominence in anticipation of noisy pricing and buyer decisions, so departures from optimal play affect both competition for sales and competition for prominence.

The experiment separates the search-cost advantage of prominence from information that rankings may convey in the field. It also shows why bids and product-market outcomes should be studied together. Auction payments depend on sellers' valuations of prominence, while those valuations depend on anticipated pricing and consumer search. An auction model that takes position values as given can describe payment incentives, but cannot alone explain how a change in consumer search friction changes those values. Conversely, a model that assigns prominence without payment misses the transfer of seller revenue to the platform.

These findings suggest several directions for further work. One concerns platforms' incentives to change search costs. In the experiment, a lower cost improves consumer and total surplus without a statistically detectable decline in auction revenue. This comparison does not establish an optimal search cost for a platform: investment in search technology may be costly, and transaction commissions, participation, and other revenue sources may also respond. An endogenous platform decision would need to incorporate these margins rather than equate platform profit with auction revenue alone.

A second question is how prominence should be allocated. Comparing the present auction with random allocation or a first-price auction would show how outcomes depend on the allocation and payment mechanisms. With more sellers and several degrees of prominence,

the allocation could also affect which products enter consumers' consideration sets. Seller differences in quality, costs, or match distributions would then matter for whether willingness to pay selects the socially preferred search order. The two identical sellers in this paper isolate the value of a search-cost advantage. They do not settle that broader allocation question.

A third question concerns the product information that sellers can display to consumers without further search. Here, prices are visible but match values must be learned through search. Other information structures may change how sellers direct attention and how buyers choose which products to search. Studying price visibility alongside product characteristics, reviews, and platform disclosure rules would help connect competition for prominence to the design of the search environment itself.

References

Abrams, E., M. Sefton, and A. Yavas (2000): "An Experimental Comparison of Two Search Models," *Economic Theory*, 16, 735–749.

Anderson, S. P. and R. Renault (1999): "Pricing, Product Diversity, and Search Costs: A Bertrand–Chamberlin–Diamond Model," *The RAND Journal of Economics*, 30, 719–735.

——— (2026): "Search Direction: Position Externalities and Position Auction Bias," *The Review of Economic Studies*, 93, 763–797.

Armstrong, M., J. Vickers, and J. Zhou (2009): "Prominence and Consumer Search," *The RAND Journal of Economics*, 40, 209–233.

Armstrong, M. and J. Zhou (2011): "Paying for Prominence," *The Economic Journal*, 121, F368–F395.

Athey, S. and G. Ellison (2011): "Position Auctions with Consumer Search," *The Quarterly Journal of Economics*, 126, 1213–1270.

Cason, T. N. and D. Friedman (2003): "Buyer Search and Price Dispersion: A Laboratory Study," *Journal of Economic Theory*, 112, 232–260.

Chen, Y. and C. He (2011): "Paid Placement: Advertising and Search on the Internet," *The Economic Journal*, 121, F309–F328.

Choi, M., A. Y. Dai, and K. Kim (2018): "Consumer Search and Price Competition," *Econometrica*, 86, 1257–1281.

Davis, D. D. and C. A. Holt (1996): "Consumer Search Costs and Market Performance," *Economic Inquiry*, 34, 133–151.

Diamond, P. A. (1971): "A Model of Price Adjustment," *Journal of Economic Theory*, 3, 156–168.

Edelman, B., M. Ostrovsky, and M. Schwarz (2007): "Internet Advertising and the Generalized Second-Price Auction: Selling Billions of Dollars Worth of Keywords," *American Economic Review*, 97, 242–259.

Goeree, J. K., C. A. Holt, and T. R. Palfrey (2020): "Stochastic Game Theory for Social Science: A Primer on Quantal Response Equilibrium," in *Handbook of Experimental Game Theory*, ed. by C. M. Capra, R. Croson, M. L. Rigdon, and T. S. Rosenblat, Edward Elgar Publishing, 8–47.

McKelvey, R. D. and T. R. Palfrey (1998): "Quantal Response Equilibria for Extensive Form Games," *Experimental Economics*, 1, 9–41.

Obradovits, M. and P. Plaickner (2023): "Price-Directed Search, Product Differentiation and Competition," *Review of Industrial Organization*, 63, 317–348.

Ursu, R. M. (2018): "The Power of Rankings: Quantifying the Effect of Rankings on Online Consumer Search and Purchase Decisions," *Marketing Science*, 37, 530–552.

Varian, H. R. (2007): "Position Auctions," *International Journal of Industrial Organization*, 25, 1163–1178.

Vragov, R., R. D. Shang, V. Smith, and D. Porter (2019): "Let's Play the Search Game: Strategic and Behavioral Properties of Sponsored Search Auction Mechanisms," *Electronic Commerce Research and Applications*, 33, 100809.

Wolinsky, A. (1986): "True Monopolistic Competition as a Result of Imperfect Information," *The Quarterly Journal of Economics*, 101, 493–511.

Xu, L., J. Chen, and A. B. Whinston (2011): "Price Competition and Endogenous Valuation in Search Advertising," *Journal of Marketing Research*, 48, 566–586.

Zhou, J. (2011): "Ordered Search in Differentiated Markets," *International Journal of Industrial Organization*, 29, 253–262.

Appendix A Omitted proofs

A.1 Proof of Lemma 1

The lemma characterizes the buyer's search and purchase decisions. At indifference, I use the following choices among optimal actions.

A1. With $v_1 = V$, the buyer stops when indifferent between searching the second seller and stopping.

A2. With $v_1 = 0$, the buyer searches the second seller when indifferent between searching and stopping.

A3. Among available choices yielding the same highest purchase surplus, the buyer prefers purchase from the top seller to purchase from the second seller, and purchase from the second seller to exit.

These rules select optimal buyer actions at equality and support the pricing equilibrium in Theorem 1. Figure 12 gives the complete buyer decision tree conditional on the observed prices and top match.

At any $p_1, p_2 \geq 0$, stopping gives $\max\{0, v_1 - p_1\}$. The expected gain from search before paying its cost is

$$q \max\{0, v_1 - p_1, V - p_2\} + (1 - q) \max\{0, v_1 - p_1, -p_2\} - \max\{0, v_1 - p_1\} \tag{24}$$

The buyer searches when this gain exceeds s and stops when it is smaller. A1 and A2 resolve equality. Once search is complete, the search cost is sunk, and A3 selects a surplus-maximizing terminal action.

For $0 \leq p_1, p_2 \leq V$ and $v_1 = 0$, the gain is $q(V - p_2)$, so A2 gives search exactly when $p_2 \leq V - \frac{s}{q}$. For $v_1 = V$, the gain is $q \max\{p_1 - p_2, 0\}$, so A1 gives search exactly when $p_1 - p_2 > \frac{s}{q}$. These prove the two cutoff rules. A price $p_1 > V$ yields no sales. If $p_2 > V - \frac{s}{q}$, the gain from search is at most $q \max\{V - p_2, 0\} < s$, so seller 2 is never searched. □

For $0 < p_1 \leq V$ and $0 < p_2 \leq V - \frac{s}{q}$, the following paths give the demand functions in (6).

(v_1, v_2)	$p_1 - p_2 > \frac{s}{q}$	$p_1 - p_2 \leq \frac{s}{q}$
$(0, 0)$	Search, then exit	Search, then exit
$(0, V)$	Search, buy 2	Search, buy 2
$(V, 0)$	Search, return to 1	Buy 1 immediately
(V, V)	Search, buy 2	Buy 1 immediately

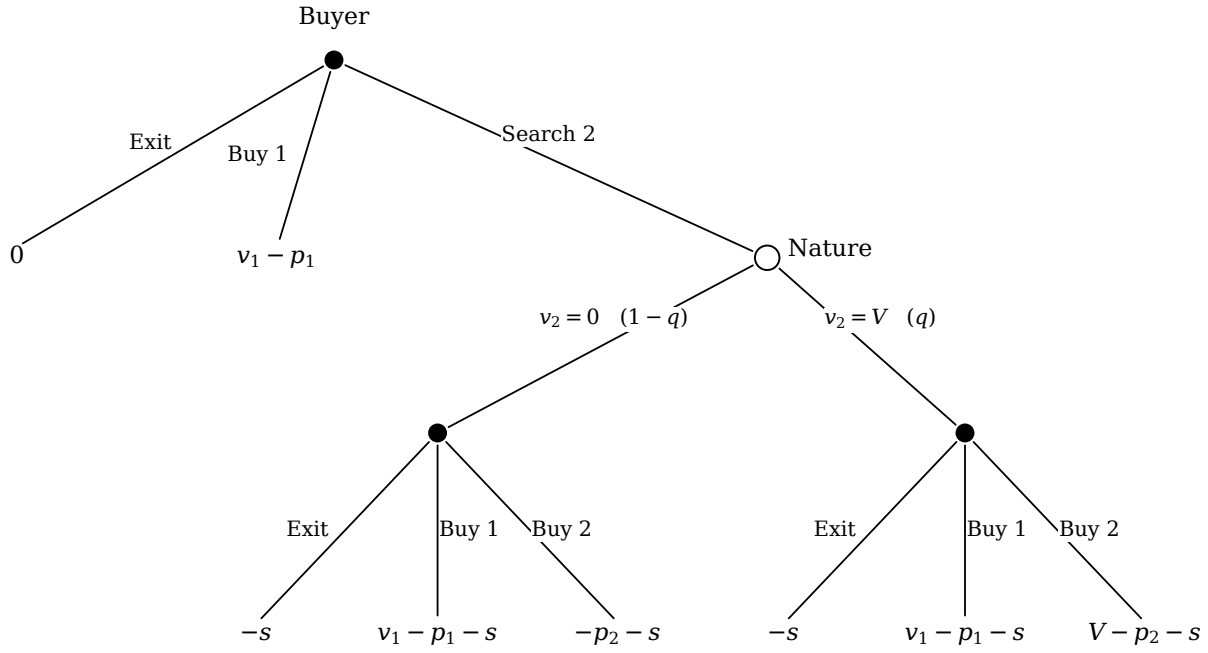


Figure 12: The buyer's complete decision tree

The buyer observes both prices and the top match before deciding whether to search the second seller. Search reveals the second match and costs s.

A.2 Proof of Proposition 1

The proposition rules out a pure-strategy equilibrium in the pricing subgame. Each seller can guarantee strictly positive revenue. Seller 1 can charge any sufficiently small $0 < p_1 < \frac{s}{q}$ and sell whenever $v_1 = V$. Seller 2 can charge any $0 < p_2 < V - \frac{s}{q}$ and sell whenever $(v_1, v_2) = (0, V)$. Thus a pure equilibrium cannot involve a zero price, $p_1 > V$, or $p_2 > V - \frac{s}{q}$.

Consider the remaining domain $0 < p_1 \leq V$, $0 < p_2 \leq V - \frac{s}{q}$. If $p_1 - p_2 > \frac{s}{q}$, seller 2 can raise its price slightly while retaining demand q. If $p_1 - p_2 < \frac{s}{q}$, seller 1 can raise its price slightly while retaining demand q. At equality, seller 2 can choose a price strictly between $(1-q)p_2$ and p_2, increasing demand from $q(1-q)$ to q and strictly increasing revenue. Every candidate price pair therefore admits a profitable deviation. □

A.3 Proof of Theorem 1

I first verify the proposed price distributions and then check each seller's possible deviations. The price bounds in Theorem 1 are

$$\underline{p}_1 = (1-q)V + s, \qquad \overline{p}_1 = V$$
$$\underline{p}_2 = (1-q)\left(V - \frac{s}{q}\right), \qquad \overline{p}_2 = V - \frac{s}{q}$$

Both proposed CDFs are increasing on their continuous parts. The first rises from zero to one. The second starts at zero and approaches $1 - \frac{s}{qV}$ at its upper bound, with the remaining mass placed at that bound. Since $0 < s < qV$, both are valid distributions with strictly positive lower bounds.

Seller 1. For $0 < p_1 \leq V$, the expected revenue against F_2 is

$$p_1 \mathbb{E}_{F_2}[D_1(p_1, p_2)] = qp_1 \left[1 - q \Pr\left(p_2 < p_1 - \frac{s}{q}\right)\right] \tag{25}$$

For $\underline{p}_1 \leq p_1 < V$, substitution of F_2 gives

$$p_1 \mathbb{E}_{F_2}[D_1(p_1, p_2)] = qp_1 \left[1 - \left(1 - \frac{(1-q)V + s}{p_1}\right)\right] = q[(1-q)V + s] \tag{26}$$

At $p_1 = V$, a good-top-match buyer stops when $p_2 = V - \frac{s}{q}$ and searches at every lower second price. Including the mass at seller 2's upper bound therefore gives

$$V\left[q\frac{s}{qV} + q(1-q)\left(1 - \frac{s}{qV}\right)\right] = q[(1-q)V + s] \tag{27}$$

A3 preserves seller 1's sales at zero purchase surplus. For $0 < p_1 < \underline{p}_1$, good-top-match buyers never search, so revenue is $qp_1 < q[(1-q)V + s]$. At $p_1 = 0$ or $p_1 > V$, revenue is zero. Hence every price in seller 1's support is a best response against all nonnegative pure-price deviations.

Seller 2. For $0 < p_2 \leq V - \frac{s}{q}$, bad-top-match buyers search. Good-top-match buyers search if $p_1 > p_2 + \frac{s}{q}$. Thus

$$p_2 \mathbb{E}_{F_1}[D_2(p_1, p_2)] = qp_2 \left[1 - qF_1\left(p_2 + \frac{s}{q}\right)\right] \tag{28}$$

For $\underline{p}_2 \leq p_2 \leq \overline{p}_2$, substituting F_1 yields

$$p_2 \mathbb{E}_{F_1}[D_2(p_1, p_2)] = qp_2 \left[1 - \left(1 - \frac{(1-q)\left(V - \frac{s}{q}\right)}{p_2}\right)\right] = q(1-q)\left(V - \frac{s}{q}\right) \tag{29}$$

This calculation includes the upper bound: A2 ensures that bad-top-match buyers still search there. For $0 < p_2 < \underline{p}_2$, both top-match types search at every price in seller 1's support, giving revenue $qp_2 < q(1-q)(V - \frac{s}{q})$. A zero price gives zero revenue, and a price above $V - \frac{s}{q}$ attracts no search. Hence every price in seller 2's support is a best response against all nonnegative pure-price deviations.

The payoff from any mixed-price deviation averages the corresponding pure-price payoffs and cannot exceed these bounds. Both optimality conditions in Definition 1 therefore hold.□

A.4 Proof of Theorem 2

The theorem derives equilibrium bids and net profits from the revenues generated by subsequent pricing. The two indifference calculations in (26) and (29) give the continuation revenues in (13). Their difference is

$$\begin{aligned}\mathbb{E}[\Pi_1] - \mathbb{E}[\Pi_2] &= q[(1-q)V + s] - q(1-q)\left(V - \frac{s}{q}\right) \\ &= qs + (1-q)s = s \end{aligned} \tag{30}$$

Against an opponent bid b, winning rather than losing changes expected net profit by $s - b$. Bidding s wins when this difference is positive, loses when it is negative, and leaves the seller indifferent at equality. It is therefore weakly dominant conditional on the specified pricing continuation. With both sellers bidding s, the position payment is s and

$$\mathbb{E}[\pi_1] = q[(1-q)V + s] - s = (1-q)(qV - s) = \mathbb{E}[\pi_2]$$

The buyer is optimal after every price pair, and Theorem 1 supplies a pricing equilibrium after every auction outcome. These strategies and the symmetric bids therefore form a subgame-perfect equilibrium. □

Appendix B Additional calculations

For each market, the purchase indicator a and search indicator x determine consumer surplus and total surplus through (16) and (17). Seller revenue is its posted price if it makes the sale and zero otherwise. Platform revenue is the losing bid. Top profit subtracts that payment from top revenue, while second profit equals second revenue. The accounting identity is checked separately for every market before treatment means are calculated. Initial balances and show-up fees are excluded from all these measures.

To test an observed outcome against its conditional equilibrium benchmark, I first calculate their difference within each market. The within-treatment comparison is a one-sample test of whether the mean difference is zero. To test whether the treatment change matches the predicted change, I compare these differences across treatments with a Welch test. This preserves the pairing between an observed outcome and the benchmark based on the same match draws.

The equilibrium price moments used below are

$$\begin{aligned} \mathbb{E}[p_1] &= \frac{s}{q} + \frac{(1-q)\left(V - \frac{s}{q}\right)}{q} \log\left(\frac{1}{1-q}\right) \\ \mathbb{E}[p_2] &= \frac{(1-q)V + s}{q} \log\left(\frac{V}{(1-q)V + s}\right) \end{aligned} \tag{31}$$

The expectations below integrate over independent equilibrium prices, conditional on the indicated match draws. The price means are given in (31). For the calculation below, the lower price bounds are

$$\underline{p}_1 = (1-q)V + s, \qquad \underline{p}_2 = (1-q)\left(V - \frac{s}{q}\right)$$

After a good top match, search occurs when $p_1 - p_2 > \frac{s}{q}$. To calculate its probability, I integrate over the independent equilibrium price distributions in (9) and (10):

$$\begin{aligned} \Pr\left(p_1 - p_2 > \frac{s}{q}\right) &= \int_{\underline{p}_1}^{V} F_2\left(p_1 - \frac{s}{q}\right) dF_1(p_1) \\ &= \frac{\underline{p}_2}{q^2} \int_{\underline{p}_1}^{V} \frac{1 - \frac{\underline{p}_1}{p_1}}{\left(p_1 - \frac{s}{q}\right)^2} dp_1 \end{aligned}$$

The atom at the second seller's upper price bound contributes no search probability, since search at $p_2 = V - \frac{s}{q}$ would require $p_1 > V$. The top seller's distribution is atomless, so the value of the integrand at $p_1 = V$ does not affect the integral.

The logarithm comes from the reciprocal terms in the decomposition

$$\frac{1-\frac{\underline{p}_1}{p_1}}{\left(p_1-\frac{s}{q}\right)^2}=\frac{q^2\underline{p}_1}{s^2}\left(\frac{1}{p_1-\frac{s}{q}}-\frac{1}{p_1}\right)+\frac{1-\frac{q\underline{p}_1}{s}}{\left(p_1-\frac{s}{q}\right)^2}$$

Integrating gives

$$\Pr\left(p_1-p_2>\frac{s}{q}\right)=\left[\frac{\underline{p}_1\underline{p}_2}{s^2}\log\left(\frac{p_1-\frac{s}{q}}{p_1}\right)-\frac{\underline{p}_2}{q^2}\frac{1-\frac{q\underline{p}_1}{s}}{p_1-\frac{s}{q}}\right]_{p_1=\underline{p}_1}^{p_1=V}$$

Using $\underline{p}_1-\frac{s}{q}=\underline{p}_2=(1-q)(V-\frac{s}{q})$, the difference between the two logarithms becomes

$$\log\left(\frac{V-\frac{s}{q}}{V}\frac{\underline{p}_1}{\underline{p}_1-\frac{s}{q}}\right)=\log\left(\frac{\underline{p}_1}{(1-q)V}\right)$$

The remaining term simplifies to $-\frac{\underline{p}_2}{s}$. Therefore,

$$\Pr\left(p_1-p_2>\frac{s}{q}\right)=\frac{\underline{p}_1\underline{p}_2}{s^2}\log\left(\frac{\underline{p}_1}{(1-q)V}\right)-\frac{\underline{p}_2}{s} \tag{32}$$

When both matches are good, expected seller revenues are

$$\mathbb{E}[\Pi_1\mid V,V]=\mathbb{E}\left[p_1\cdot\mathbf{1}_{\{p_1-p_2\le\frac{s}{q}\}}\right]=\frac{(1-q)V+s-(1-q)\mathbb{E}[p_1]}{q} \tag{33}$$

$$\mathbb{E}[\Pi_2\mid V,V]=\mathbb{E}\left[p_2\cdot\mathbf{1}_{\{p_1-p_2>\frac{s}{q}\}}\right]=\frac{(1-q)\left(V-\frac{s}{q}\right)-(1-q)\mathbb{E}[p_2]}{q} \tag{34}$$

These identities follow by conditioning on the seller's own price and substituting the other seller's CDF. With $(v_1,v_2)=(0,V)$, only seller 2 makes a sale and earns p_2. With $(V,0)$, only seller 1 makes a sale and earns p_1. With $(0,0)$, neither makes a sale. A bad-top-match buyer always searches at equilibrium prices, whereas a good-top-match buyer searches with the probability in (32). Applying the surplus identities in (16) and (17) gives Table 7.

Table 7: Conditional equilibrium outcomes and observed match frequencies

s	(v_1, v_2)	Count	$\mathbb{E}[\Pi_1]$	$\mathbb{E}[\Pi_2]$	$\mathbb{E}[\text{CS}]$	$\mathbb{E}[\text{TS}]$
20	$(0, 0)$	97	0.00	0.00	-20.00	-20.00
20	$(0, 100)$	119	0.00	49.93	30.07	80.00
20	$(100, 0)$	105	81.59	0.00	13.08	94.67
20	$(100, 100)$	99	58.41	10.07	26.19	94.67
40	$(0, 0)$	112	0.00	0.00	-40.00	-40.00
40	$(0, 100)$	104	0.00	18.96	41.04	60.00
40	$(100, 0)$	113	93.86	0.00	2.91	96.77
40	$(100, 100)$	121	86.14	1.04	9.60	96.77

Expectations are conditional on each match pair. Counts sum to 420 and 450. Platform revenue equals s in every row, top profit is top revenue minus s, and second profit equals second revenue. Treatment benchmarks weight the rows by their observed frequencies. Values are rounded for display.

Let ρ denote the probability of search after a good top match in (32). A buyer with a bad top match always searches at equilibrium prices. Averaging over the initial match therefore gives

$$\mathbb{E}[x] = (1 - q) + q\rho \tag{35}$$

At equilibrium, a product with value V is purchased whenever at least one match is good. The probability is $1 - (1 - q)^2 = q(2 - q)$. Hence

$$\mathbb{E}[\text{TS}] = q(2 - q)V - s[(1 - q) + q\rho] \tag{36}$$

$$\mathbb{E}[\text{CS}] = \mathbb{E}[\text{TS}] - \mathbb{E}[\Pi_1] - \mathbb{E}[\Pi_2] \tag{37}$$

These formulas yield the unconditional values in Section 3.3. The empirical benchmarks instead weight the conditional outcomes in Table 7 by the observed match frequencies. The buyer's search decision in either calculation depends only on the initial match and posted prices, not the second match before search.

Appendix C Experimental instructions[16]

Welcome! Thanks for participating in this experiment on economic decision-making. Your participation is completely voluntary. Please remain seated and do not use your cellphone. You will interact anonymously through the computers. Please do not talk or look at others' screen.

At the end of the experiment, we will record your earnings amount and you will be paid by cash privately.

Fixed Role and Random Group

In this experiment, you will play the role of either a **Seller** or a **Buyer**. Your role will be assigned at the beginning and will remain fixed throughout the entire experiment.

There are **30 rounds** in total. In each round, you will be randomly grouped into a market consisting of **3 participants** (including yourself): **2 Sellers and 1 Buyer**. Groups are **randomly shuffled** every round, so you will likely interact with different people in the room from round to round. You will never know exactly who you are matched with.

Products and Fit Values

Sellers offer products to the Buyer. A seller's product can either be a Good Fit or a Bad Fit for the buyer, with equal chances.

- **Good Fit:** Value to the buyer is **100 points**. This happens with a chance of **50%**.
- **Bad Fit:** Value to the buyer is **0 points**. This happens with a chance of **50%**.

Before visiting a seller, the buyer does NOT know whether the product will be a good fit or bad fit. The fit is determined **randomly** and **independently** for each seller. The buyer only knows a product's fit value ***after visiting*** that seller. Knowing the fit with one seller does not provide any information about the fit with the other seller.

Decisions in Three Stages

Seller Bidding $\Longrightarrow$ Seller Pricing $\Longrightarrow$ Buyer Visit & Purchase

Each round proceeds according to the following three stages:

Stage I: Sellers Bid and Pay for Top Position

The two sellers simultaneously submit bids (in points) to compete for the "Top Position" on the Buyer's screen. A seller cannot see the other seller's bid when choosing their own bid. Bids must be between 0 to 100 points.

- **The Top Seller:** The seller with the higher bid wins and is displayed **first** to the buyer. It costs the buyer **0 points** to visit the **Top Seller**.
- **The Second Seller:** The seller with the lower bid loses and is displayed **second** to the buyer. It costs the buyer **20 points** to visit the **Second Seller**. The buyer decides whether or not to pay this cost, meaning the Second Seller might not be visited at all.

[16] These instructions are for the low-search-cost treatment. In the high-search-cost treatment, the search cost is 40 rather than 20 experimental points.

Tie-breaking: If both sellers submit the exact same bid, the computer will randomly select one seller to win and become the Top Seller (each seller has an equal chance).

How the Bidding Payment Works

The **Top Seller** wins the position, but pays a position payment equal to the **Second Seller's bid** (not their own bid). The **Second Seller** pays **0 points**.
Example: Seller A bids **22**. Seller B bids **14**. Seller A becomes the Top Seller, but Seller A only pays **14 points** (not 22). Seller B becomes the Second Seller and pays **0 points**.

Stage II: Sellers Set Prices Observed by Buyer

After sellers see whether they are the Top Seller or Second Seller, they simultaneously post a price for their product. Once posted, both prices are **immediately observed** by the buyer. When setting prices, sellers do not know if their product is a Good Fit or Bad Fit for the buyer.

Stage III: Buyer Visits and Purchases

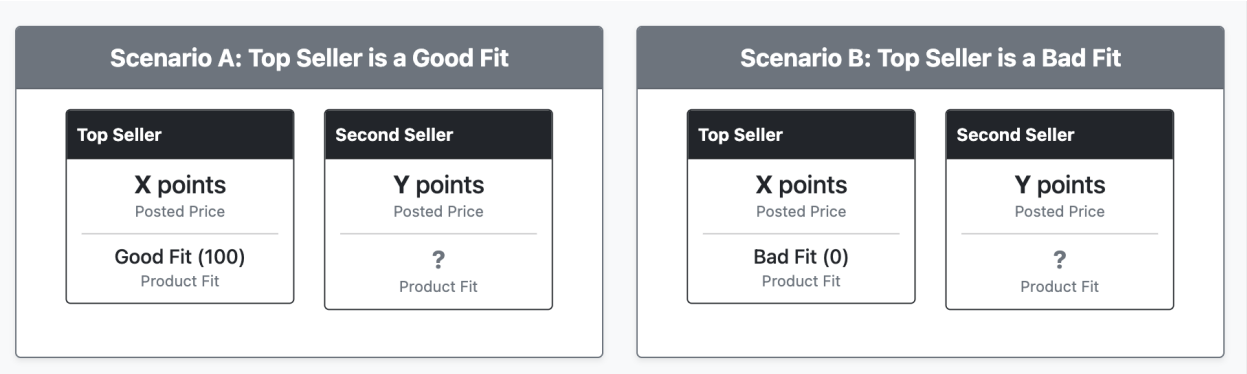
The buyer observes the prices posted by both sellers. The buyer automatically visits the **Top Seller** for free and learns its product fit.

The buyer then decides whether to pay a **Visit Cost** of **20 points** to discover the fit of the **Second Seller**. **The Visit Cost is non-refundable**, even if the buyer ultimately decides not to buy from the Second Seller after the visit.

Finally, the buyer can choose to purchase one product from any seller the buyer has visited. Specifically, a buyer who visits the Second Seller can still choose to buy from the Top Seller, without paying any *additional* visit cost. The buyer may also choose to quit the market at any time and buy nothing.

Examples of Buyer's Decision Screen

For example, if sellers post prices of **X** and **Y**, the buyer observes **one** of these two screens, depending on whether Top Seller is randomly determined to have a Good Fit or Bad Fit. Notice that buyer always knows Top Seller's fit, but Second Seller's fit remains **?** until visited.



Points and Earnings

At the end of the experiment, the computer will randomly select **ONE out of the 30 rounds** to be your "Payment Round." Your points in this Payment Round will be converted into dollar earnings using a conversion ratio of **10 points = $1**.

You will start each round with an initial balance of **200 points** (worth $20), which will be added to your payoff points in that round. This starting balance resets each round and does not carry over. Your total earnings for this experiment will be **$10** show-up fee plus your converted points in your (randomly selected) Payment Round.

Your payoff points are calculated as follows:

Buyer Payoff Points:

- If you purchase: **Product Fit** − **Price Paid** − **Visit Cost** (if you visit Second Seller)
- If you quit without purchase: − **Visit Cost** (if you visit Second Seller)

Notice that Visit Cost is non-refundable for the buyer.

Seller Payoff Points:

- If you make a sale: **Price Charged** − **Position Payment**
- If you do not make a sale: − **Position Payment**

Notice that Position Payment is non-refundable for sellers.

For each round: **Total Points = 200 Points + Payoff Points**

Please be patient while waiting for others to finish their decisions. At the end of each round, you will see a feedback screen showing your payoff and total points for that round.

Below is a summary of the instructions. Next, you will answer three comprehension questions and play two practice rounds. These practice rounds help you learn the interface and do not affect your earnings. After that, we will start the formal rounds. Once you have made a choice, you cannot go back to a previous screen to change it. **If you have any questions, please raise your hand at any time during the experiment.**

Instructions Summary

Rounds	**30 rounds**. Your role is fixed: you are either a **Seller** or a **Buyer**.
Groups	Each round: **2 Sellers + 1 Buyer** are **randomly** grouped.
Three Stages	Bidding $\Longrightarrow$ Pricing $\Longrightarrow$ Visit & Purchase
Bidding	Highest bid becomes **Top Seller** and pays the other Seller's bid.
Pricing	Prices posted by both Sellers are **observed** by Buyer before any visit.
Product Fit	When Buyer visits a Seller, Buyer discovers if the product is a **good fit** worth **100 points** (**50%** chance), or a **bad fit** worth **0 points** (**50%** chance). The fit is determined **independently** for each seller.
Visit Cost	Visiting **Top Seller** is free. Visiting **Second Seller** costs Buyer **20 points**.
$Earnings	Show-up **$10** + Points of **ONE** random round (**10 points = $1**).

Appendix D Screenshots

This appendix presents screenshots of the experimental interface for the high-search-cost treatment ($s = 40$).[17]

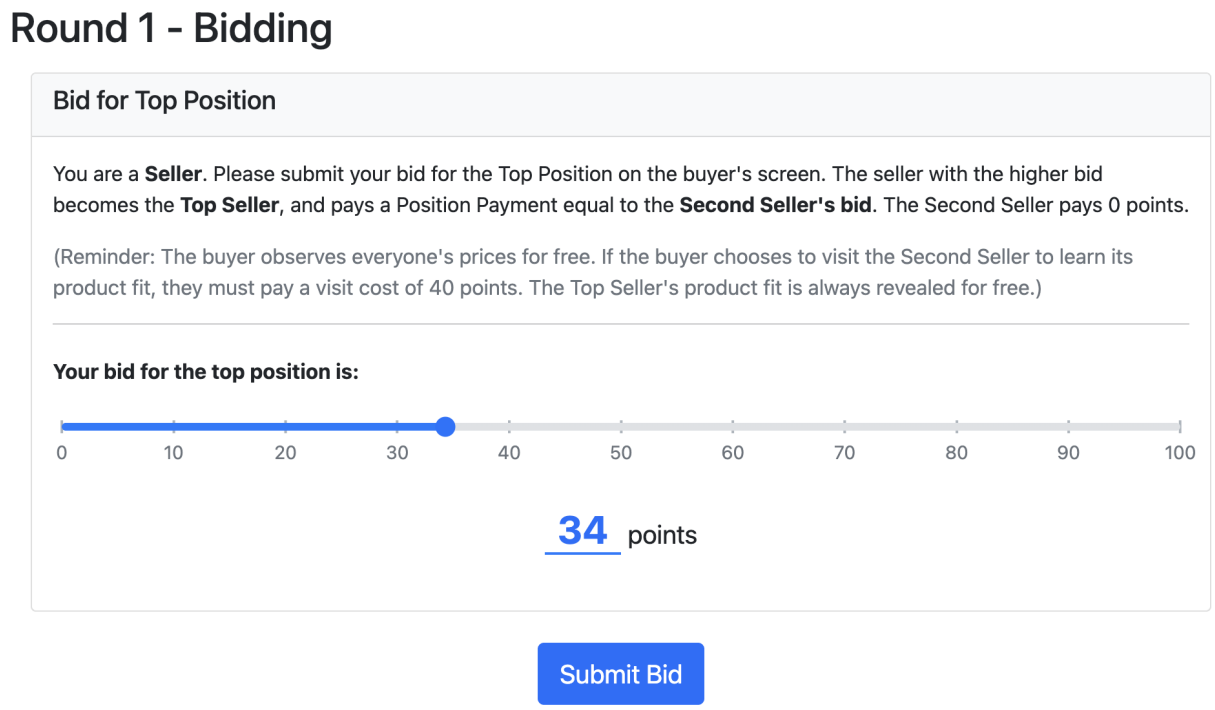


Figure 13: Seller's Bidding Decision

[17]The screens and instructions are identical in the low-search-cost treatment ($s = 20$), except that the search cost is displayed as 20 rather than 40 experimental points.

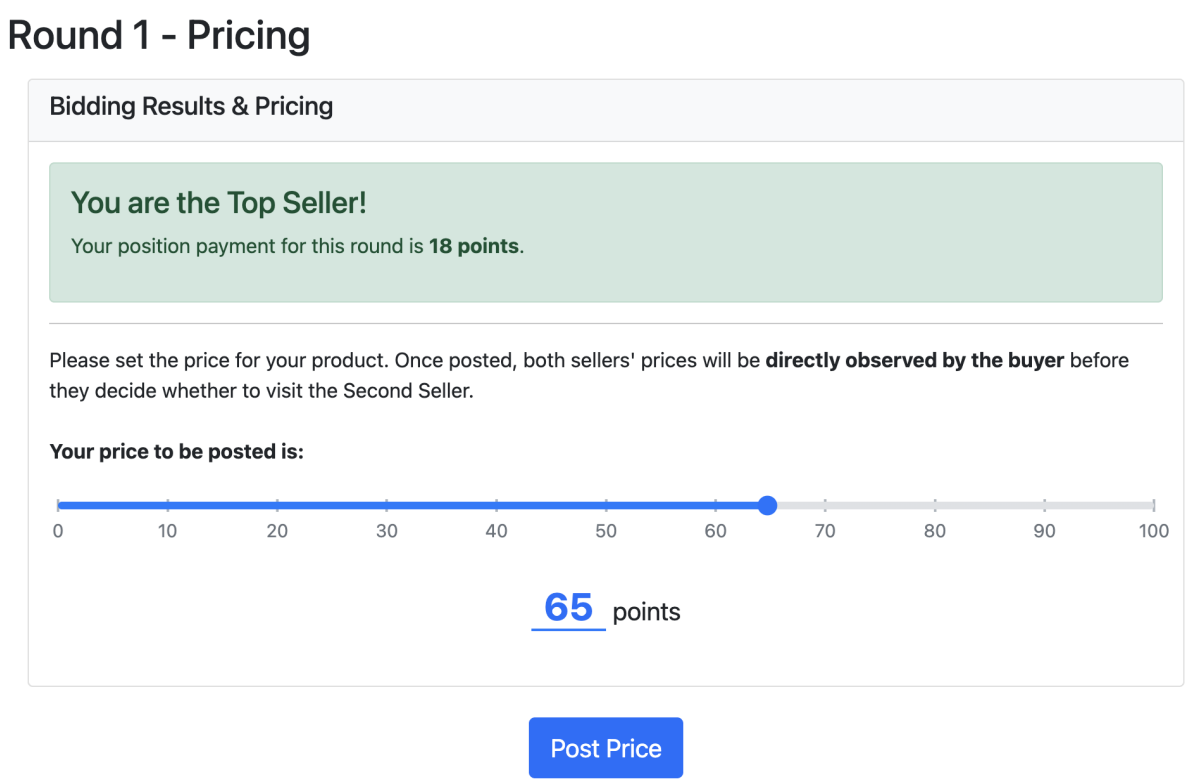


Figure 14: Top Seller's Pricing Decision

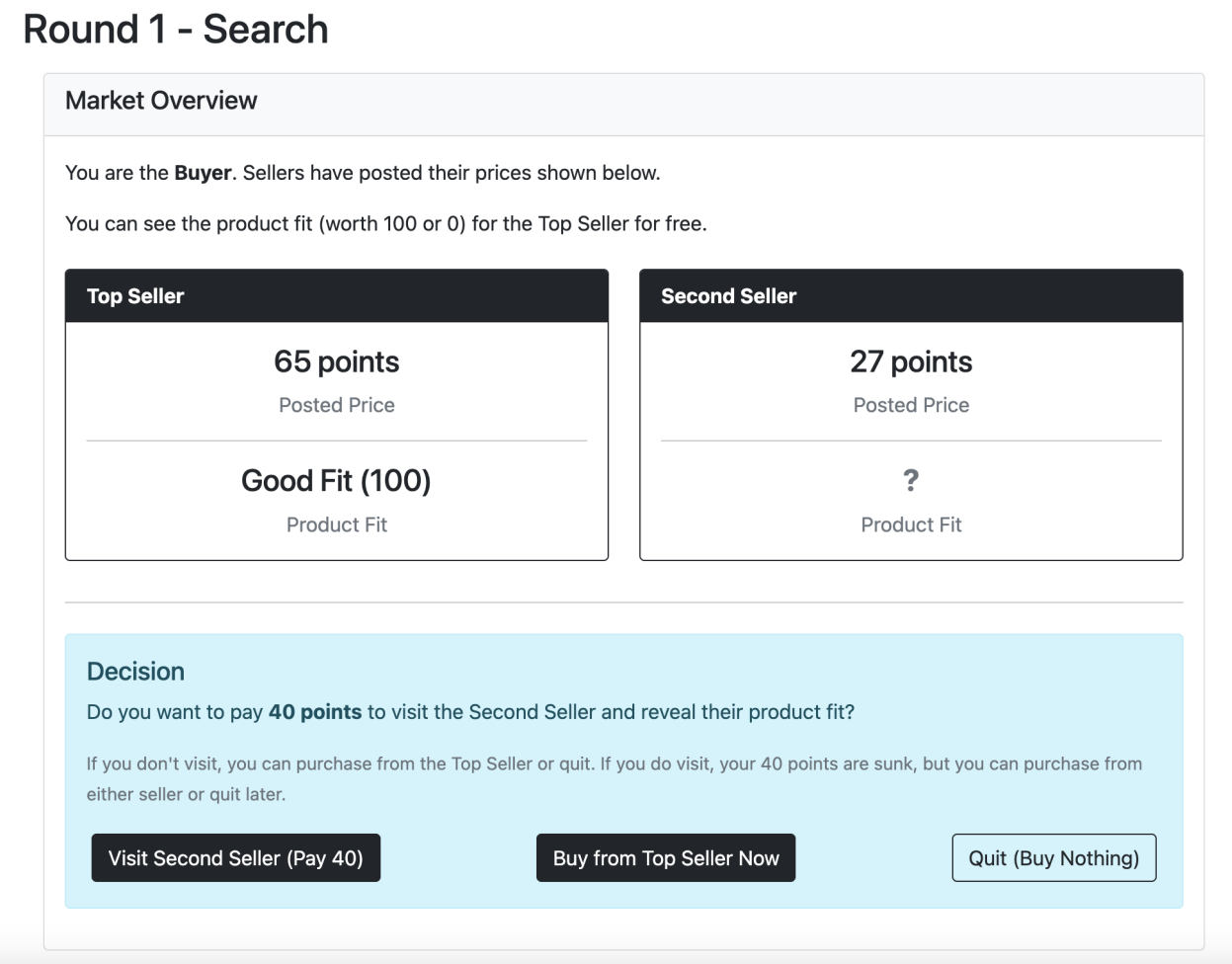


Figure 15: Buyer's Search Decision

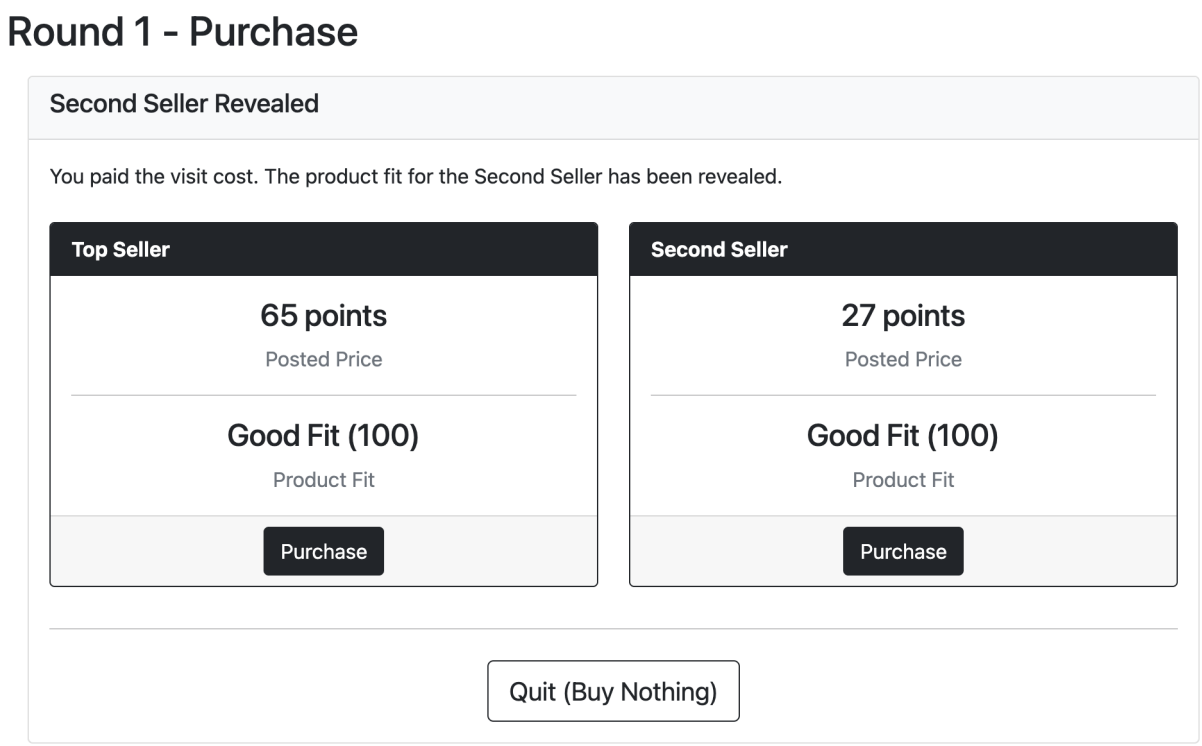


Figure 16: Buyer's Purchase Decision after Search

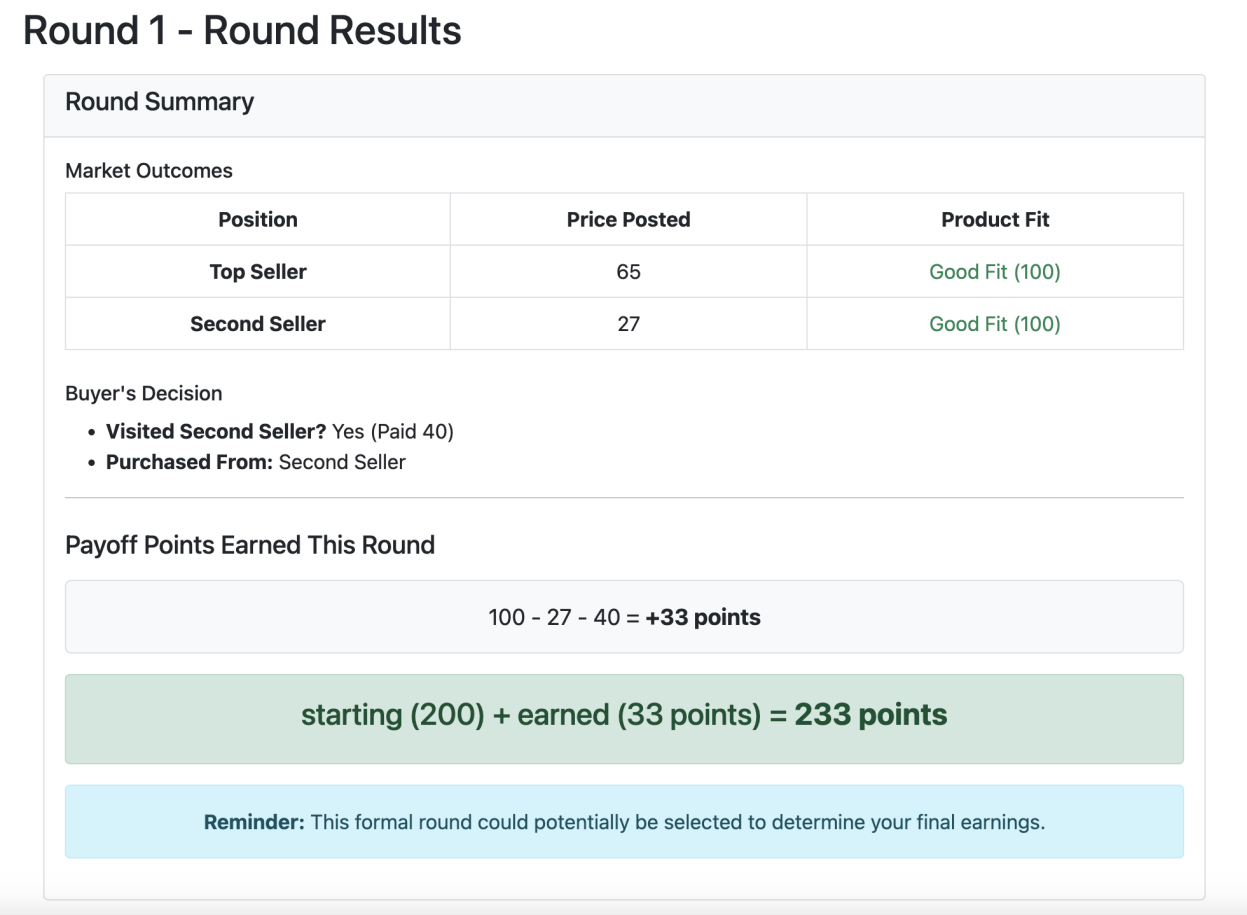


Figure 17: Buyer's Round-End Feedback